Journal of Financial Economics 144 (2022) 247–272

Contents lists available at ScienceDirect

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/jfec

Network effects in corporate financial policies[☆]

William Grieser[a], Charles Hadlock[b,*], James LeSage[c], Morad Zekhnini[b]

[a] Texas Christian University, United States
[b] Michigan State University, United States
[c] Texas State University, United States

ARTICLE INFO

Article history:
Received 9 November 2020
Revised 28 April 2021
Accepted 1 May 2021
Available online 2 June 2021

JEL classification:
G31
G32
C40

Keywords:
Capital structure
Peer effects
Networks
Spatial econometrics

ABSTRACT

We present a spatial econometrics framework for estimating peer effects in capital structure. This approach exploits the heterogeneous and intransitive nature of peer networks to identify economically informative structural coefficients. In models of leverage levels, we detect significant peer-effect leverage coefficients that are on the order of 0.20, indicating a moderate but substantive level of strategic complementarity in capital structure decisions. We argue that prior estimates in the literature substantially overstate the magnitude of the underlying relation. Our evidence is robust to a wide variety of model modifications and supports the hypothesis that leverage is an important strategic choice variable.

1. Introduction

Most firms operate in environments in which the actions of product-market peers can have a significant effect on the firm's profitability. If firms account for these effects, many corporate policy choices should represent equilibrium outcomes that incorporate feedback effects across product-market rivals. To explore this issue, several prominent theoretical studies model the interrelated capital structure policies of firms within a product-market environment. Empirical investigations offer support for elements of these theories in specific industry settings.

Despite the potential importance of this general issue, our understanding of the role of peer effects in capital structure policies for broad sets of firms is limited. In an important paper that helps fill this gap, Leary and Roberts (2014) examine the role of peer average capital structure as a determinant of a given firm's capital structure choice. They present evidence indicating the presence of causal peer effects by exploiting a source of exogenous variation in peer capital structure choices. While their study represents an important step forward, the economic magnitude of the underlying peer-effect relation is not cleanly identified by their study, owing to a variant of the Manski (1993) reflection problem. As we discuss, exogenous variation alone is not sufficient for identifying the magnitude of peer effects using traditional regression techniques, as the evolution of equilibrium outcomes is nuanced in the presence of interrelated firm decisions. Evidently, quantifying the economic magnitude of peer effects in capital structure decisions requires a richer modeling approach.

☆ We thank an anonymous referee for valuable comments and suggestions. We also thank Gerard Hoberg, Paul Irvine, Bill Johnson, Swami Kalpathy, Sandy Klasa (discussant), Gordon Phillips, Bill Schwert (the editor), Shawn Thomas, and Jeff Wooldridge, as well as seminar participants at the 2019 Finance Organizations and Markets Conference, Suffolk University, Texas A&M University, and Tulane University. All errors remain our own.

* Corresponding author.
E-mail address: hadlock@msu.edu (C. Hadlock).

https://doi.org/10.1016/j.jfineco.2021.05.060
0304-405X/

In this paper, we offer economically meaningful estimates of the magnitude of peer effects in capital structure by exploiting recent developments both in the measurement of product-market network structure and in spatial econometrics techniques. Using the approach pioneered by Hoberg and Phillips (2010, 2016), we map out the network structure of a firm's product-market peers and use the heterogeneous and intransitive nature of these networks to capture the relative strength of peer interactions. Importantly, this allows us to exploit richer information regarding variation in product-market industry structures to assist in identification. Recent developments in spatial econometrics techniques (see, e.g., Lee, 2007; Bramoullé et al., 2009) then allow us to econometrically model equilibrium capital structure choices when firms' decisions heterogeneously affect different industry peers, in turn yielding estimates of structural parameter values.

Since we utilize two relatively new methodological elements in our study, we attempt to maintain consistency with the prior literature, particularly Leary and Roberts (2014, LR hereafter), in our other modeling choices. Thus, our baseline models utilize the same set of covariates selected by LR, with an emphasis on the equity shock variable that they highlight as an exogenous external driver of peer capital structure choice. After establishing results using baseline choices, we experiment with alternatives. Most notably, inspired by Almeida et al. (2012), we exploit an alternative exogenous external driver of capital structure based on debt that is soon coming due. Our findings with regard to peer effects using this alternative confirm our initial evidence, indicating that our findings are not highly sensitive to the particular choice of exogenous variation used to identify the coefficients of interest.

Our findings indicate the presence of significant causal peer effects in capital structure decisions. While there are subtleties in interpreting the findings, the baseline estimates indicate an initial sensitivity of a firm's leverage choice to peer leverage on the order of 0.20. This strikes us as an economically reasonable figure and suggests a prominent, but not overwhelming, role for peer capital structure choices on the decisions of any given firm. The positive sign for this relation indicates that leverage choices behave as strategic complements across the product-market network. Importantly, our framework allows the overall effect of an exogenous shifter in leverage choices for one firm on other firms to depend on the location of a firm in the network.

Our estimates do differ substantially from the much higher figures that appear in the prior literature, many of which we find implausibly large. For example, LR report a close to one-for-one implied change in firm leverage in response to a change in peer average leverage. When we investigate these differences, we find that estimates derived from prior approaches can be quite fragile, possibly because they do not correspond directly to the structural coefficient of interest. We also find that the assignment rule for identifying peer networks is quite important, with assignments based on the Hoberg and Phillips (2010; 2016) product similarity scores leading to models with a substantially better fit than traditional Standard Industrial Classification (SIC) based approaches.

The associated prior literature is mostly silent on the economic mechanism underlying detected peer effects, with strategic, learning, and managerial channels all possibly playing a role. We argue the empirical modeling approach we adopt is most likely to detect strategic effects, suggesting that our evidence primarily reflects the role of competitive considerations in which firms choose a capital structure that is substantively influenced by the role of that structure in the firm's product-market game. To further explore the economic mechanism underlying our findings, we consider variations in the strength of estimated peer effects across different subsamples of firms. Here, we provide some modest evidence of larger peer effects in settings in which some economic channels might predict an enhanced role for peers in leverage decisions.

While we focus on capital structure decisions, the methodology we employ is likely to be useful for understanding other firm choices in which theoretical considerations suggest substantive peer interfirm interactions. For example, a firm's investment decisions will likely depend in an important way on peer firms' decisions (and vice versa) (see Dougal et al., 2015; Bustamante and Frésard, 2020). While fully investigating these possibilities is beyond the scope of our paper, we offer preliminary evidence on the potential for our modeling approach to identify peer effects in other corporate finance settings. This analysis also suggests that the approach we advance may be useful in noncorporate contexts, for example, the role of social networks on investor behavior (see, e.g., Heimer, 2016; Cookson et al., 2021; Kuchler et al., 2020; Bailey et al., 2019).

An additional observation that emerges from our analysis of different settings is that estimated coefficients on nonpeer variables can change substantially when peer effects are included in the estimated models. This suggests that allowing for peer effects could be an important robustness check in settings in which the role of peers is not the question of primary interest. Our evidence along these lines echoes the point of LeSage and Pace (2009) that allowing for peer effects can, in some cases, ameliorate omitted variable bias in coefficient estimates on nonpeer variables.

The rest of the paper is organized as follows. In Section 2, we discuss the prior literature and outline the methodological approach we adopt to overcome challenges inherent in many prior studies. In Section 3, we outline our sampling procedures and characterize the sample. Our main results on peer network effects in capital structure decisions are reported in Section 4. In Section 5, we consider extensions of our approach to related questions. Section 6 concludes.

2. Motivation and empirical strategy

2.1. Channels for peer capital structure effects

An extensive theoretical literature explores the strategic role of capital structure in product-market behavior.[1] This

[1] Seminal early models include Brander and Lewis (1986), Maksimovic (1988), and Bolton and Scharfstein (1990). Other promi-

stream of research illustrates that the presence of risky debt can affect equityholders' returns to different product-market strategy choices and in turn it also affects competitors' choices. The conclusions from this theoretical literature with regard to ex ante capital structure decisions are typically quite model specific.[2] However, collectively they provide convincing motivation for the general hypothesis that strategic considerations will lead firms to respond to peer leverage choices as part of their own capital structure strategy. Depending on the precise theory, the sign of the relation could be positive (strategic complements) or negative (strategic substitutes).

In addition to strategic channels, prior work suggests the possibility of learning elements to capital structure decisions in which firms make inferences on optimal choices by observing the decisions and experiences of industry peers. Of note, the widely observed importance of industry effects in capital structure policy suggests the presence of an industry-related component to many of the parameters that govern optimal choices (see, e.g., Bradley et al., 1984; Frank and Goyal, 2009). If this is indeed the case, optimal learning rules may include a dynamic response of any given firm's decision to its peers' choices.

In a different vein, research in other contexts suggests the possibility of managerial-related motivations for peer effects in capital structure. In particular, in the spirit of the Scharfstein and Stein (1990) model, managers' career incentives to herd may drive firms to mimic peers, even when managers are aware that they are deviating from a value-maximizing strategy. Alternatively, managers may be influenced by behavioral rules that rely heavily on specific peers' choices when making inferences about optimal capital structures.

2.2. Prior empirical work and identification challenges

A growing empirical literature explores peer effects in a variety of finance contexts including dividend policy (Adhikari and Agrawal, 2018; Grennan, 2019), stock splits (Kaustia and Rantala, 2015), capital investment (Dougal et al., 2015; Bustamante and Frésard, 2020; Cookson, 2018), corporate social responsibility policies (Cao et al., 2019), and bank liquidity choice (Silva, 2019).[3] Many of these papers are agnostic on whether the peer effects they uncover are related to strategic or nonstrategic considerations. In some cases, a certain interpretation is emphasized, often based on the scenario that applies most and/or that best explains a constellation of findings.

Turning to the specific issue of capital structure, several early empirical papers offer evidence supporting elements of theories regarding strategic considerations in specific industry contexts (see, e.g., Chevalier, 1995; Phillips, 1995; Zingales, 1998).[4] Other authors use broad general samples to examine the cross-sectional variation in leverage as a function of the product-market environment (see, e.g., MacKay and Phillips, 2005; Almazan and Molina, 2005). This latter and more general evidence appears broadly consistent with the widespread presence of strategic effects in the determination of equilibrium capital structure choices. However, while these investigations uncover interesting patterns, the picture they present is incomplete, owing to the inability of traditional regressions to adequately capture within-industry interactions.

More recently, Leary and Roberts (2014) make a significant step forward by modeling a firm's capital structure as a function of peer average leverage. Using a novel instrument for peer choices, they present instrumental variable (IV) evidence of a causal role for the peer average. As LR discuss, a major challenge in interpreting studies that use peer averages as explanatory variables arises from the well-known Manski (1993) reflection problem. The main issue is that peer averages should have an effect on the dependent variable, and also be affected by the dependent variable, leading to a vexing simultaneity identification challenge. Unfortunately, the usual methods that exploit an external instrument for identification are not adequate in this context, as IV estimates will represent the effect of interest only up to an unidentified scaling factor (details below). Thus, even when a defensible instrument is available and the null hypothesis of no peer effects can be rejected, as in LR, coefficient magnitudes have no clear interpretation. Despite this acknowledged limitation, many authors often offer an economic interpretation of estimated coefficient magnitudes. Our evidence below suggests that these interpretations can be misleading.

An important component of the LR study is their focus on exogenous variation in peer firm capital structures catalyzed by idiosyncratic components of equity returns. This outside information helps substantially in identification, as many sources of endogeneity of the omitted-variable flavor can be eliminated.[5] Not surprisingly, several other peer effects studies have followed the LR lead in using exogenous sources of variation based on idiosyncratic shocks to components of equity returns (see, e.g., Adhikari and Agrawal, 2018; Grennan, 2019; Fairhurst and Nam, 2020). By combining this important LR innovation with richer information on industry network structure, recent advances in spatial econometrics can be used to infer economically meaningful estimates of the magnitude of causal peer effects.[6]

An attractive feature of the LR approach of exploiting publicly observed idiosyncratic shocks is that the underlying variation is particularly well suited for isolating strategic peer effects. To illustrate, note that in a strategic scenario, when one firm's policy choice, say

nent models include Showalter (1995), Chevalier and Sharfstein (1996), and Dasgupta and Titman (1998).

[2] See the surveys by Maksimovic (1995) and Parsons and Titman (2008).

[3] In addition, several studies study the role of peer effects in the context of managerial networks. See, for example, Shue (2013) and Fracassi (2016).

[4] Related prominent empirical studies include Kovenock and Phillips (1997), Khanna and Tice (2000, 2005), and Campello (2003; 2006).

[5] See Angrist (2014) for a vivid illustration of these sources of endogeneity in peer-effects studies in labor economics.

[6] Silva (2019) adopts a related approach in a study of strategic bank liquidity choices by exploiting both exogenous instruments and variation in network structure.

capital structure, is exogenously perturbed, this will start a causal reaction through the network as firms strategically re-optimize their leverage choices in the now perturbed product-market game. In a learning scenario, idiosyncratic shocks for one firm should not contain useful technological (i.e., nonstrategic) information for another firm's optimal choice, so reactive changes in leverage should not be evident absent strategic feedback effects. Similarly, many managerial career concerns theories rely on the role of unobserved private signals influencing a decision to mimic peers. Thus, public idiosyncratic signals should be largely irrelevant, as outside market participants (e.g., the labor market) should fully adjust to neutralize this information in the signal extraction process.

2.3. Exploiting industry network structure

The straightforward approach of using a peer average based on traditional industry classifications as an explanatory variable in predicting a firm's policy choice implicitly assumes that all peer firms should carry equal weight in influencing a firm's decision. In addition, it assumes that firms operate in a group structure in which each firm is equally related to all other firms within the group and unrelated to all firms outside of the group. Clearly, these features do not capture the rich real-world heterogeneity in the degree to which firms interact by virtue of variation in product mix, geographic location, customer base, etc. By ignoring this important variation, models that utilize the equal-weighted peer average approach will be unnecessarily coarse. Moreover, as we discuss below, the assumed symmetry of interactions in this approach often leaves key coefficients unidentified (or very weakly identified), even in the presence of exogenous variation.

To more realistically model peer effects and achieve convincing identification, a natural approach is to incorporate into the modeling a measure of the degree to which two firms interact and compete in the product market. Fortunately, influential work by Hoberg and Phillips (2010, 2016) offers an intuitive and validated off-the-shelf method to accomplish this goal. Using textual analytics applied to product-market discussions in a firm's SEC filings, these authors offer a measure of both a firm's industry peer network and the degree to which two firms in the network sell similar products or services.[7]

As a starting point for modeling the strength of peer interactions, we build on the Hoberg and Phillips (2016) product similarity scores, which range in value from 0 to 1, as recorded in their text-based network of industry competitors (TNIC) database. To limit attention to relatively close competitors, for any firm i and peer j in a given year, we set s_{ij} equal to the similarity measure for firm-pair i, j if firm j is a top 10 peer according to this score, and 0 otherwise. For each firm i, we normalize these scores to create a weighted measure w_{ij}, where w_{ij} is set equal to s_{ij} divided by the sum of all s_{ij} across j ($i \neq j$, year t held fixed). If there are n firms in year t, the associated $n \times n$ weighting matrix with w_{ij} as the entry in row i and column j will be referred to as the annual peer-weighting matrix W_t. This matrix will have diagonal values of 0 and row sums of 1.

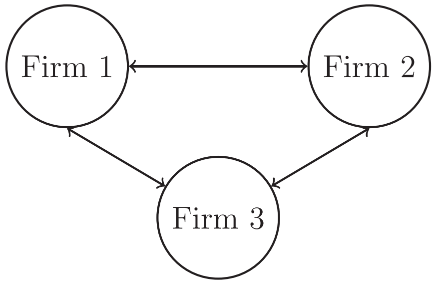

Fig. 1. A three-firm transitive network.

If we combine the annual peer-weighting matrices across years to form a block-diagonal matrix (with w_{ijt} denoting the entry in row i, column j, block t), we arrive at an overall sample peer-weighting matrix. This matrix, which we denote by W, will serve as our default/baseline peer network characterization in the analysis that follows. While we expect this choice of W to be a substantial improvement over many other choices, we recognize that it still may only imperfectly capture the true nature of actual interfirm interactions. To provide insights on the potential impact of this measurement error issue on our conclusions, we explore below the sensitivity of our findings to variations in the assumed structure of this matrix.

2.4. Identification of peer effects

Armed with a peer-weighting matrix, spatial econometric techniques can be applied to estimate peer effects in capital structure choices. Given the novelty of this estimation approach in corporate finance settings, we first illustrate the variation in the data that is exploited to overcome identification limitations in related prior studies. This also allows us to highlight the nuanced issue of identifying the magnitude of peer effects, even in contexts in which the presence of nonzero effects can be convincingly established. The key intuition can be obtained in a simple example in which we assume the availability of data for many industries, each composed of three firms.

2.4.1. Transitive network case

We start by considering what is commonly referred to as a transitive network (i.e., situations in which peers of peers are always peers). Specifically, we consider the situation depicted in Fig. 1 in which, for each three-firm industry in a given year, firms interact in a symmetric manner.

If we denote an outcome variable for firm i in industry k, say capital structure, by y_i, the symmetry of this network leads naturally to a model in which each firm's choice depends on the average of its two competitors' choices, with a magnitude captured by a parameter ρ,

[7] We implicitly assume that underlying firm interactions that lead to peer capital structure effects are closely related to levels of product similarity. While this appears intuitively reasonable, it is not a direct implication of the underlying theories which are highly stylized. In some of our analysis, we experiment with different choices for characterizing the relevant peer network.

where $-1 < \rho < 1$. Allowing x_i to be a covariate that directly affects only a firm's own policy choices, and allowing an industry constant, leads to the following data generating process:

$$\begin{cases} y_1 = \alpha_k + \rho\frac{y_2+y_3}{2} + \beta x_1 + \epsilon_1 \\ y_2 = \alpha_k + \rho\frac{y_1+y_3}{2} + \beta x_2 + \epsilon_2 \\ y_3 = \alpha_k + \rho\frac{y_1+y_2}{2} + \beta x_3 + \epsilon_3 \end{cases} \quad (1)$$

As is well known in this setting, an OLS regression across industries of each firm's dependent variable on the average of its peers' values (along with explanatory variable x) will lead to inconsistent parameter estimates, as the peer average variable and the error terms will clearly be correlated. One can solve the system to express the endogenous y variables as a function of the exogenous x variables to arrive at a reduced form equation for each firm i of:

$$y_i = \frac{\rho}{(1-\rho)(2+\rho)}\sum_j(\alpha_k + \beta x_j + \epsilon_j) + \frac{2}{2+\rho}(\alpha_k + \beta x_i + \epsilon_i),\ i = 1, 2, 3. \quad (2)$$

To eliminate the industry-specific constant, we let $\ddot{y}_i$ ($\ddot{x}_i$, $\ddot{\epsilon}_i$) represent y_i (x_i, ϵ_i) less the corresponding average quantity for the firm's two industry peers. Simple algebra then yields:

$$\ddot{y}_i = \frac{\beta}{1+\rho/2}\ddot{x}_i + \frac{1}{1+\rho/2}\ddot{\epsilon}_i,\ i = 1, 2, 3. \quad (3)$$

This equation illustrates one of the key identification issues highlighted by Manski (1993), namely that the role of a covariate (β) and the peer average (ρ) on the dependent variable cannot be separately identified, even if we had data from firms in many industries of the same size (i.e., many sets of 3 firm industries). If we allow a given firm's choice to depend on the covariates of other firms (so called contextual effects), the problem is further complicated.

To intuitively illustrate the identification challenge and the underlying economics, consider a small perturbation of x_1 by δ. Given the data generating process, the initial result will be an increase in y_1 of $\beta\delta$, followed by an initial peer-effect increase in y_2 and y_3 of $\beta\delta\rho/2$. As y_2 increases, this will have a feedback effect on y_3, and vice versa, and they will iterate through a geometric series process resulting in a total change in y_2 and y_3 (and their average) of $(\beta\delta\rho/2)/(1-\rho/2)$. This can be viewed as the end of round 1. Round 2 will then commence with Firm 1 altering y_1 by an amount equal to ρ times the first-round change in y_2 and y_3, followed by another set of proportional changes in y_2 and y_3. This process will continue to cycle through rounds that can be represented as a converging geometric series (provided that $\rho < 1$). The reduced form in (3) represents the ultimate impact of this entire set of adjustments.

The identification challenge is readily apparent from this thought exercise. The initial perturbation of x_1 results in a change in y_1 that is proportional to β. This in turn increases y_2 and y_3 (and thus their average) by an amount proportional to β times an increasing function of ρ, owing to the peer effect. Thus, when we consider the industry-demeaned change in y_1 (i.e., $\ddot{y}_1$), a moderate direct effect (moderate β) with a negligible peer effect (small or zero ρ) can often look identical to a larger direct effect (larger β) that is offset in the demeaning by a larger peer effect (a larger ρ). All future rounds involve factors that are proportional to ρ, and thus this intuition carries over with slight modification to the final demeaned change in y_1 as a result of a perturbation in x_1. Similar logic applies also to all of the cross-partials (i.e., $\partial\ddot{y}_i/\partial\ddot{x}_j$ for $i \neq j$).

2.4.2. Intransitive network case

The transitive case, which is implicitly assumed in almost all prior related studies, leads to symmetries in the reduced form equations that result in the identification challenges discussed above. However, there is clearly a great deal of intransitivity and variation in real-world product-market interactions, and it is precisely these features that the peer-weighting matrix W discussed above based on the Hoberg and Phillips (2016) similarity scores captures. As an example, Nike competed for years with Callaway in golf and with Head in tennis, but Callaway and Head did not compete directly with each other.

When a network is intransitive, the resulting reduced form explaining the equilibrium value of the dependent variables as a function of the exogenous variables results in many more equations because of a lack of symmetry, and this, in turn, results in more information that can be exploited for identification. Intuitively, if peer effects are negligible, then a shock to an exogenous variable should have a similar effect on policy choices at Nike, Head, and Calloway. However, if peer effects are substantial, shocks to Nike may have a relatively larger effect on Head and Callaway relative to the cross-firm transmission of other shocks, given Nike's more central position in the network. It is this variation in network position that we exploit in estimation.[8]

To illustrate more formally, consider an alternative three-firm network depicted in Fig. 2 in which one firm (Firm 2) is central to the network, and the other two firms only directly interact with this central firm.

In this case, if peer effects are present, it is reasonable to expect Firm 1 and 3's policy choices (the dependent variable y) to directly depend only on Firm 2, while Firm 2's choices could directly depend on the average of the other two firms. These assumptions are captured by the following data generating process:

$$\begin{cases} y_1 = \alpha_k + \rho y_2 + \beta x_1 + \epsilon_1 \\ y_2 = \alpha_k + \rho\frac{y_1+y_3}{2} + \beta x_2 + \epsilon_2 \\ y_3 = \alpha_k + \rho y_2 + \beta x_3 + \epsilon_3. \end{cases} \quad (4)$$

Solving for the reduced form equations for the y terms and subtracting peer averages to eliminate any industry

[8] The probability that a peer of a peer is also a direct peer is 0.299 in our product-market network. For a transitive network (e.g., defined based on SIC codes), the probability would be 1.0.

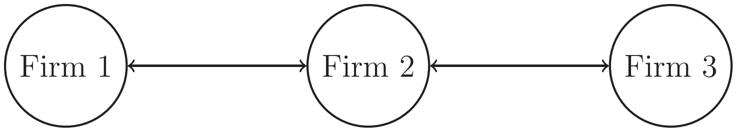

Fig. 2. A three-firm intransitive network.

constant α_k for industry k yields:

$$\begin{cases} \ddot{y}_1 = \frac{1}{1-\rho^2}\Big[\beta\Big(\ddot{x}_1 + \frac{\rho}{3}(\ddot{x}_2 - \frac{\ddot{x}_1+\ddot{x}_3}{2}) + \frac{\rho^2}{2}(\ddot{x}_3 - \ddot{x}_1)\Big) \\ \qquad + \ddot{\epsilon}_1 + \frac{\rho}{3}(\ddot{\epsilon}_2 - \frac{\ddot{\epsilon}_1+\ddot{\epsilon}_3}{2}) + \frac{\rho^2}{2}(\ddot{\epsilon}_3 - \ddot{\epsilon}_1)\Big] \\ \ddot{y}_2 = \frac{1}{1-\rho^2}\Big[\beta\Big(\ddot{x}_2 - \frac{2\rho}{3}(\ddot{x}_2 - \frac{\ddot{x}_1+\ddot{x}_3}{2})\Big) + \ddot{\epsilon}_2 - \frac{2\rho}{3}(\ddot{\epsilon}_2 - \frac{\ddot{\epsilon}_1+\ddot{\epsilon}_3}{2})\Big] \\ \ddot{y}_3 = \frac{1}{1-\rho^2}\Big[\beta\Big(\ddot{x}_3 + \frac{\rho}{3}(\ddot{x}_2 - \frac{\ddot{x}_1+\ddot{x}_3}{2}) + \frac{\rho^2}{2}(\ddot{x}_1 - \ddot{x}_3)\Big) \\ \qquad + \ddot{\epsilon}_3 + \frac{\rho}{3}(\ddot{\epsilon}_2 - \frac{\ddot{\epsilon}_1+\ddot{\epsilon}_3}{2}) + \frac{\rho^2}{2}(\ddot{\epsilon}_1 - \ddot{\epsilon}_3)\Big]. \end{cases} \tag{5}$$

As is evident, the intransitive structure results in equations that are not symmetric, as Firm 2's sensitivity to variation in the covariates will differ from the other two firms because of its more central location in the network.[9] It is immediately apparent from this set of equations that the ratio of any two distinct derivatives of y values to x values will eliminate β by cancellation, allowing the parameter ρ (and thus also β) to be recovered. Thus, we see that an intransitive network allows asymmetries in sensitivities to be exploited to recover the structural parameters of interest.

2.4.3. The use of exogenous variation

It is tempting to maintain the assumption of symmetric/transitive interactions across peers and to try to solve the identification challenge by identifying an instrument for peer leverage followed by an application of standard Two-Stage Least Squares (2SLS) estimation techniques. Inspired by Duflo and Saez (2003), this is the approach followed by LR.[10] However, our discussion above maintains the assumption that the model covariates (i.e., the x variables) are exogenous, yet identification of the peer-effect coefficient ρ in the transitive case is not readily apparent. Thus, if one estimates a 2SLS regression with exogenous variables serving as instruments for the peer average of the dependent variable, it must be the case that the resulting coefficient cannot capture solely the structural peer-effect coefficient of interest (ρ).

To clarify these issues, in our analysis below, we revisit parts of the LR evidence, including estimating versions of their 2SLS models. While LR interpret the resulting 2SLS coefficients as an estimate of ρ, we emphasize that it is unclear what 2SLS estimated coefficients on a peer average represent, as these coefficients will be a function of both ρ and other model parameters. Exogenous variation in a selected explanatory variable may very well initially impact only a peer (or the peer average), as is necessary to serve as an instrument. However, the initial impact will then be transmitted to other firms via the peer effect, and then back to the original firm(s) via the peer of a peer effect (and so on). Thus, even with clean and informative exogenous variation of covariates, the instrumented values from a first-stage regression of the peer average in a 2SLS framework cannot possibly satisfy the exclusion restriction, essentially by construction. The approach we advocate overcomes this problem by combining asymmetries in the peer network structure with exogenous variation to isolate the peer-effect parameter of interest.

2.4.4. Additional identification issues

In the simple example outlined above, the covariate x directly affects only a firm's own dependent variable choice. It is certainly possible for a given firm's covariates to also directly affect peer choices, a possibility that is usually referred to as a contextual effect. As discussed by Manski (1993) and Leary and Roberts (2014), if these effects are represented by some parameter θ, it will typically be infeasible to identify θ in the transitive case. However, a straightforward extension of the preceding three-firm example with contextual effects reveals that θ is identified in the intransitive case.

Before proceeding with the empirical analysis, we emphasize that a causal interpretation of the underlying parameters of interest, ρ and β (and θ, if allowed to be nonzero), still relies on the usual assumption that the explanatory variables (i.e., the x variables) are, in fact, exogenous. However, if these variables have some endogenous elements, it is sometimes the case that a causal β coefficient can be identified even if ρ no longer has a causal interpretation (see LeSage and Pace, 2009, for details). Essentially, the adjustment for common shocks via allowing for peer effects may be sufficient to capture omitted variable bias so that a causal β can be recovered. Thus, an adjustment for peer effects could at times serve as a useful robustness check when the coefficients on other (nonpeer) explanatory variables are of primary interest.

2.5. Applying spatial econometrics techniques

Spatial econometrics techniques, which were developed to model the role of activity in one region on an adjoining region, offer a general estimation strategy for the matrix analog of the examples presented above. The key parameters of interest are ρ, which as above represents the magnitude of the peer-effect relation, a coefficient vector β, measuring the direct role of explanatory variables

[9] As in the transitive case, the derivatives implied by these equations can be intuitively derived by considering a small perturbation in an exogenous variable and then tracing out the subsequent rounds of feedback effects which are captured by a set of converging geometric series.

[10] Both LR and Duflo and Saez (2003) refer to an earlier Case and Katz (1991) working paper to justify their 2SLS estimation strategies. However, Case and Katz (1991) discuss the benefits of a rudimentary version of the spatial econometrics approach.

on a given firm's dependent variable choice, and, in some models, θ, a vector measuring the role of other firms' explanatory variables on a given firm's choice. The choice of whether to allow θ to be nonzero is unclear, as few related models posit a direct role for peer explanatory variables in a firm's dependent variable. However, there are advantages of letting the data speak, and thus we experiment with both treatments.

In what follows, we let y denote the dependent variable (i.e., capital structure) values for a set of firm-years, W the peer-network weighting matrix described earlier characterizing the strength of individual peer interactions, and X a matrix of explanatory variable values, one row corresponding to each firm-year. If we assume no contextual effects (i.e., $\theta = 0$), we have what is commonly referred to as the Spatial Autoregressive (SAR) model with the relationship described by:

$$y = \rho W y + X\beta + \epsilon. \tag{6}$$

If we allow for contextual effects, we have the augmented model commonly labeled the spatial Durbin model (SDM) with the relationship given by:

$$y = \rho W y + X\beta + WX\theta + \epsilon. \tag{7}$$

Clearly, the SAR and SDM structural equations cannot be estimated directly, and thus we must first solve for their reduced forms in terms of y. These reduced forms for the SAR and SDM models, respectively, are given by:

$$y = (I - \rho W)^{-1}(\beta X + \epsilon) \tag{8}$$

and

$$y = (I - \rho W)^{-1}(X\beta + WX\theta + \epsilon). \tag{9}$$

These equations are nonlinear in the parameters, and thus a nonlinear estimation method is necessary. Consequently, we follow the common strategy in these settings of conducting an MLE estimation that exploits Markov Chain Monte Carlo (MCMC) methods. Technical details regarding model estimation and associated inference issues are relegated to the Appendix. Importantly, the use of a peer-weighting matrix W that incorporates an intransitive peer network structure assures that these estimates will be uniquely identified, as intuitively illustrated in the example earlier.

2.6. Interpreting estimates

Before turning to the data, we discuss the interpretation and economic content of parameter estimates derived from this framework. If a given explanatory variable x_k for firm i is perturbed, this will shift firm i's dependent variable choice. In the presence of nonzero peer effects as described by the SAR or SDM models, other firms should choose new capital structures in response, which in turn will be reflected back to the original firm (firm i), and also to other peers (i.e., peers of peers). The ultimate impact of the resulting feedback loops on the new equilibrium, implicitly assumed to take place within the periodicity of the data, can be of a substantially different magnitude than the actual estimated parameters. Since the weighting matrix W allows each firm to be linked to its peers in a different way based on product similarity scores, the final impact of a perturbation in any explanatory variable on dependent variable values will vary for each firm.

In our discussions of peer effects, we at times refer to the peer-effect coefficient ρ as an "initial" effect, followed by subsequent reactions/reverberations through the peer network, as we find this to be a useful way to conceptualize the underlying re-equilibration process. However, we emphasize that we do not explicitly consider any dynamics regarding how firm capital structures re-equilibrate. It may be that the process is a series of reactions, and then reactions to reactions, and so on, all taking place in real time within the annual observation window. Alternatively, it may be that a new equilibrium described by the equations and intuition above occurs more-or-less simultaneously (as implied by a period-by-period static game-theoretic perspective). Whatever the case, we are effectively modeling the annual level re-equilibration process whenever there is an external force on capital structure at one firm that is then transferred to other firms through the system, via the peer-effect coefficient ρ, to arrive at a new equilibrium.[11] In the special case of a symmetric set of equal-weighted peers, the geometric-series characterization of the re-equilibration yields a total effect of ρ divided by $1 - \rho$.

Turning to the interpretation of the other model covariates, we follow LeSage and Pace (2009) and calculate average direct and indirect effects arising from perturbations in the exogenous explanatory variables. The average direct effect for covariate x_k is defined to be the estimated impact of a one-unit change in this variable for firm i on firm i's dependent variable choice (i.e., capital structure) in the new equilibrium, averaged over all firms and years. Similarly, the average indirect effect is the estimated cumulative impact of this same change in the explanatory variable x_k at firm i on the dependent variable choices of all other firms (except firm i), again averaged over all firm-years. The coefficient on ρ will clearly be a key input into these estimated effects, as it captures (in a geometric series fashion) the magnitude of each step of the transmission of the initial perturbation through the peer network. Importantly, the ρ coefficient has a causal interpretation in the sense that it represents a structural feedback effect from a peer's choice to a firm's own choice. However, it is not economically useful if viewed in isolation, as the (implicit) equilibrium perspective we adopt does not allow for an isolated change in dependent variable choices.

3. Data and sample selection

To aid in comparisons, many of our empirical choices follow the treatment of prior authors. All accounting and market information is derived from the CRSP-Compustat merged database, and most variables are constructed to match the treatment outlined by LR. For dependent variables, we use the same set of six capital structure policy variables studied by LR. Following their treatment, we

[11] For practical reasons, we follow the prior literature and empirically model the equilibration process at the annual level. Thus, we do not allow for additional lagged effects from dynamic adjustments.

Table 1

Summary statistics. This table provides summary statistics for the primary variables used in the analysis. The variables are constructed using the methodology of Leary and Roberts (2014). *Book Leverage* is the ratio of total debt to book assets, *Market Leverage* is the ratio of total debt to the market value of assets, *Log(Sales)* is the natural log of sales, *Market − to − Book* is the ratio of market equity to book equity, *EBITDA/Assets* is profitability, and *Net PPE/Assets* is tangibility. Following Leary and Roberts (2014), the *EquityShock* variable is based on lagged idiosyncratic stock returns measured as the residual from a return model that includes market and industry factors. *Debt Coming Due* is the proportion of long-term debt coming due within one year. The table summarizes these variables in levels (first three columns) and year-over-year first differences (last three columns). For each variable, we report statistics for the average of all firms in the same three-digit SIC code as the focal firm (excluding the focal firms) as well as the own firm characteristics. The sample period is 1996 to 2017.

	Levels			First differences		
	Mean	Median	SD	Mean	Median	SD
Peer firm averages						
Book leverage	0.233	0.205	0.113	0.011	0.010	0.040
Market leverage	0.214	0.176	0.129	0.008	0.007	0.048
Log(sales)	5.591	5.512	1.208	0.056	0.065	0.118
Market-to-book	1.706	1.537	0.727	−0.022	−0.009	0.367
EBITDA/assets	0.054	0.080	0.105	−0.008	−0.004	0.047
Net PPE/assets	0.255	0.189	0.182	−0.002	−0.002	0.016
Equity shock	0.009	−0.024	0.273			
Firm-specific factors						
Book leverage	0.223	0.178	0.225	0.009	0.000	0.100
Market leverage	0.213	0.135	0.235	0.008	0.000	0.112
Log(sales)	5.599	5.670	2.221	0.057	0.060	0.297
Market-to-book	1.645	1.207	1.352	−0.027	−0.003	0.853
EBITDA/assets	0.066	0.110	0.203	−0.006	0.000	0.116
Net PPE/assets	0.254	0.182	0.225	−0.002	−0.002	0.048
Equity shock	0.009	−0.122	0.858			
Debt coming due	0.164	0.056	0.248			
Industry characteristics						
Number of firms per industry-year	14.015	7	25.764			
Total number of industries	200					
Sample characteristics						
Observations	46,433					
Firms	6132					

predict these variables as a function of four standard explanatory variables (measures of size, growth opportunities, profitability, and tangibility), plus the novel idiosyncratic equity shock variable that LR highlight as an exogenous shifter of capital structure choices. This variable, which has been used in several recent related finance studies, captures the difference between realized and predicted equity returns derived from a regression model. LR report that these shocks are associated with subsequent decreases in leverage levels and increases in equity financing. Specific details regarding variable constructions are reported in the Appendix (see Table A.1).

We select as the sample period 1996–2017, a choice that is dictated by the availability of the Hoberg and Phillips (2010, 2016) product similarity scores that are needed for characterizing the peer network. As is common in this literature, we exclude utilities (SIC codes 4900–4999) and financial firms (SIC codes 6000–6999). All non-dummy variables are winsorized at the upper and lower 1% tails. Our final sample includes 6132 firms from 1996 to 2017. Table 1 reports basic summary statistics for the sample.

4. Empirical results

4.1. Initial evidence on peer effects

We first estimate the simplest possible peer-effects model in our framework, which we refer to as the parsimonious model. In this model, we include the equity shock variable as the only exogenous explanatory variable. An advantage of this model is that any endogeneity in the control variables will not contaminate the key coefficient estimates of interest, although this may come at the cost of estimation precision. We initially assume the SAR model structure in (6) above in which a firm's leverage choice in a given year is assumed to be determined by the peer parameter ρ times the weighted average of peer choices (weighted by peer similarity for the ten closest peers as represented in matrix W), plus some coefficient β times the firm's equity shock, plus noise. We estimate the reduced form (8) above using MLE implemented by MCMC methods, and then recover the structural parameter values from the reduced form estimates. Following LR, we allow for industry and year fixed effects in all baseline models.

Table 2

Baseline models of peer effects in corporate leverage decisions. The table presents estimates of coefficients/effects for the parsimonious Spatial Autoregressive (SAR) model outlined in the text and Appendix in which corporate leverage decisions are assumed to be a function of a weighted average of peer leverage and an additional exogenous explanatory variable. Standard errors are reported in brackets under each estimate. In Panel A, the exogenous variable is a firm's lagged idiosyncratic equity shocks. In Panel B, the exogenous variable is the fraction of the firm's long-term debt that is due in the coming year. The table includes both the actual estimated coefficients on the exogenous variables and their direct and indirect effects as described in the text. The parameter ρ is the peer-effect coefficient for the estimated SAR model. Peers are identified and weighted using the product-market similarity measure of Hoberg and Phillips (2016) as represented by the peer-network matrix W outlined in the text. Variables are normalized by their annual standard deviations before the estimation and all estimated models include industry and year fixed effects, where industries are defined as the Fama-French 48 industries. The estimation procedure is Maximum Likelihood implemented with Markov Chain Monte Carlo (MCMC) methods and detailed in the Appendix. The six dependent variables capturing capital structure policy are selected and constructed to be consistent with Leary and Roberts (2014). The sample includes 41,549 CRSP-Compustat firm-years with available observation from 1996 to 2017.

	Market leverage	Book leverage	Equity issuance	Debt issuance	Δ Market leverage	Δ Book leverage
Panel A. Equity shock exogenous variable						
Peer-effect coefficient ρ	0.250***	0.184***	0.135***	0.061***	0.113***	0.039***
	[0.004]	[0.004]	[0.004]	[0.005]	[0.004]	[0.004]
Equity shock	−0.090***	−0.047***	0.109***	−0.007	−0.069***	−0.057***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Direct effects						
Equity shock	−0.091***	−0.048***	0.109***	−0.007	−0.070***	−0.057***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Indirect effects						
Equity shock	−0.029***	−0.010***	0.017***	0.000	−0.009***	−0.002***
	[0.002]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]
Panel B. Debt due exogenous variable						
Peer-effect coefficient ρ	0.257***	0.135***	0.165***	0.067***	0.142***	0.042***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Debt due	−0.174***	−0.190***	0.069***	−0.078***	−0.018***	−0.014***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Direct effects						
Debt due	−0.177***	−0.190***	0.069***	−0.078***	−0.018***	−0.014***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Indirect effects						
Debt due	−0.058***	−0.029***	0.013***	−0.006***	−0.003***	−0.001***
	[0.002]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]

Significance at * 10% level, ** 5% level, *** 1% level

We report estimates for models explaining the six dependent variables in the columns of Panel A of Table 2. We first report the actual coefficient estimates, namely the peer-effect coefficients ρ, and the equity shock coefficients β. Since these coefficients by themselves do not convey the complete economic picture, we also report the estimated total direct and indirect effects of a perturbation in the equity shock variable. As discussed earlier, these effects effectively represent, respectively, the average net effect of a perturbation of the exogenous explanatory variable on the firm itself, and on all other firms, after accounting for the convergence of all feedback effects through the product-market network. Clearly, larger peer effects, as indicated by a large peer-effect ρ coefficient, will tend to magnify feedback effects.

The reported ρ coefficients in the first row of Panel A of Table 2 provide strong evidence of significant peer effects in capital structure decisions. The coefficients are positive and significant in all models. Since many prior authors focus on modeling capital structure levels (see, e.g., Titman and Wessels, 1988), we largely focus on the economic content of the first two columns, which are for models that predict a firm's level of market and book leverage, respectively. The highly significant peer-effect coefficients here of 0.250 and 0.184, respectively, indicate that an exogenous 1% increase in the weighted average of peer leverage will result in an initial increase in a firm's leverage of around one-fifth of that amount. This initial increase will then be transmitted to other firms in the network, and in turn, also back to the original firm, so the total change will be larger than this amount (on the order of ρ / $1 - \rho$), with a converging geometric series feature to the process.

It is difficult to make more precise statements on the economic magnitude of the peer effect, as ideally one needs to model an underlying cause of a peer leverage perturbation. That is exactly what the equity shock variable does. A shift of this variable at a peer will tend to exogenously perturb the peer's capital structure (with a strength proportional to β), which in turn will be transmitted to other firms in the network, with a strength of transmission measured by ρ times the weight measuring the rel-

Table 3

Extended models of peer effects in corporate leverage decisions. The table presents estimates of coefficients/effects for the extended Spatial Autoregressive (SAR) model outlined in the text and Appendix in which corporate leverage decisions are assumed to be a function of a weighted average of peer leverage, a selected exogenous explanatory variable, and a set of four additional control variables. Standard errors are reported in brackets under each estimate. In Panel A, the selected exogenous variable is a firm's lagged idiosyncratic equity shocks. In Panel B, the selected exogenous variable is the fraction of the firm's long-term debt that is due in the coming year. The reported estimates for the exogenous and control variables are the direct and indirect effects as outlined in the text, rather than actual coefficient estimates. The parameter ρ is the peer-effect coefficient for the estimated SAR model. Peers are identified and weighted using the product-market similarity measure of Hoberg and Phillips (2016) as represented by the peer-network matrix W outlined in the text. Variables are normalized by their annual standard deviations before the estimation and all estimated models include industry and year fixed effects. All other model, sample, and estimation details are as described in the note to Table 2.

	Market leverage	Book leverage	Equity issuance	Debt issuance	ΔMarket leverage	ΔBook leverage	Market leverage	Book leverage	Equity issuance	Debt issuance	ΔMarket leverage	ΔBook leverage
	Panel A. Equity shock exogenous variable						Panel B. Debt due exogenous variable					
Peer-effect coefficient ρ	0.194***	0.152***	0.083***	0.052***	0.094***	0.031***	0.223***	0.128***	0.129***	0.064***	0.130***	0.028***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Direct effects												
Equity shock	−0.045***	−0.023***	0.104***	0.009*	−0.064***	−0.053***	−0.164***	−0.193***	0.019***	−0.065***	-0.020***	-0.016***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.005]	[0.004]	[0.004]	[0.004]	[0.005]	[0.004]	[0.004]
Log(sales)	0.128***	0.189***	−0.121***	0.133***	0.088***	0.078***	0.020***	0.071***	−0.173***	0.047***	0.052***	0.039***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]
Market-to-book	−0.284***	−0.035***	0.221***	−0.027***	−0.220***	−0.007	−0.262***	0.053***	0.158***	-0.014***	-0.164***	0.060***
	[0.004]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.004]	[0.005]	[0.005]	[0.005]	[0.004]
Profit	−0.176***	−0.196***	−0.132***	−0.117***	−0.195***	−0.206***	−0.174***	−0.305***	−0.009*	-0.052***	-0.151***	-0.363***
	[0.005]	[0.005]	[0.005]	[0.006]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Tangibility	0.166***	0.191***	−0.002	0.095***	0.052***	0.074***	0.104***	0.101***	−0.034***	0.059***	0.076***	0.059***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.004]	[0.004]	[0.005]	[0.004]	[0.004]
Indirect effects												
Equity shock	−0.010***	−0.004***	0.009***	0.000*	−0.006***	−0.002***	−0.045***	−0.028***	0.003***	-0.004***	-0.003***	0.000***
	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]	[0.002]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]
Log(sales)	0.030***	0.033***	−0.011***	0.007***	0.009***	0.002***	0.006***	0.010***	−0.025***	0.003***	0.008***	0.001***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]
Market-to-book	−0.066***	−0.006***	0.020***	−0.001***	−0.022***	0.000***	−0.072***	0.008***	0.023***	-0.001***	-0.024***	0.002***
	[0.002]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]	[0.002]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]
Profit	−0.041***	−0.034***	−0.012***	−0.006***	−0.020***	−0.007***	−0.048***	−0.043***	−0.001*	-0.003***	-0.022***	-0.010***
	[0.002]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.002]	[0.002]	[0.001]	[0.000]	[0.001]	[0.001]
Tangibility	0.039***	0.033***	0.000	0.005***	0.005***	0.002***	0.029***	0.014***	−0.005***	0.004***	0.011***	0.002***
	[0.001]	[0.001]	[0.000]	[0.001]	[0.001]	[0.000]	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]	[0.000]

Significance at * 10% level, ** 5% level, *** 1% level ative connection between firms. As we report in the second row of Panel A of Table 2, the estimated coefficients on β are all significant and of the expected sign (i.e., as in LR, positive (negative) in all models predicting more (less) leverage), with the exception of debt issuance.[12] The direct effect estimates, reported in the third row, are quite similar to the β coefficient estimates, while the indirect effects reported in the subsequent row are, as expected, substantially smaller.

The parsimonious SAR model in Panel A of Table 2 does not allow for contextual effects in which an equity shock at one firm directly affects leverage decisions at peer firms. Some authors allow for this additional channel, and thus we have also estimated a corresponding SDM model (see (7) above) in which we add a parameter θ that is multiplied by the weighted average of peer equity shocks as an additional determinant of leverage. The estimates from this model, which are omitted to conserve space, are quite similar to what we report in Table 2. In particular, the estimated peer-effect parameter ρ and the β parameter are essentially unchanged in these more general models.

4.2. Evidence from an alternative exogenous determinant of capital structure

As in LR, a causal interpretation of the preceding findings does depend on the exogeneity of the explanatory variables, and one may be unconvinced that the equity shock variable satisfies this assumption. Thus, it would be comforting to continue to find evidence for the presence of peer effects by using an alternative variable to serve as a plausibly exogenous force in capital structure determination. To investigate, we consider variation in capital structure arising from a firm's debt maturity structure. In particular, we exploit the observation of Almeida et al. (2012) that there is likely a substantial exogenous component to the portion of a firm's long-term debt that is soon due, as the maturity decision for this debt was often made many years in the past. Since debt coming due must be either paid off or refinanced, frictions or fixed costs in the process of refinancing may result in some of the maturing debt being paid off, thus tending to lower a firm's leverage ratio.

Following Almeida et al. (2012), we define this variable, *debtdue*, as the firm's long-term debt due in one year, divided by total long-term debt, all measured as of the start of the observation year. We expect the coefficient on this variable to generally be negative in predicting leverage. Assuming that this variable is exogenous, the estimated ρ coefficients in SAR or SDM models including this variable can be viewed as alternative estimates of peer effects. If these ρ estimates are comparable to the earlier estimates, it strengthens the case for a causal/structural interpretation of the findings.

We present estimates from models that use this variable in place of the equity shock variable in Panel B of Table 2. As the figures in the first row reveal, the estimated peer-effect coefficients (ρ) are similar to the corresponding estimates from the initial models. They all remain positive and highly significant, and the only change of any notable magnitude is in the book leverage model (column 2) in which the estimate on the peer-effect parameter drops somewhat from 0.184 to 0.135. The coefficient estimates reported in the second row for the *debtdue* variable are all significant and of the expected sign. These findings are substantively unchanged if we consider the SDM analogs of the tabulated SAR models.

4.3. Robustness of initial evidence

The findings in Table 2 provide strong initial evidence of peer effects in capital structure. The positive signs on the estimated effects indicate strategic complementarity in leverage decisions, and the coefficients in the leverage level models suggest that a firm responds to peer weighted-average leverage choices with a sensitivity on the order of 0.20. In this subsection, we consider the robustness of these findings.

The parsimonious models discussed above exploit variation in a single exogenous variable. As we report, the case for peer effects is quite strong from these models, so that the usual compelling need to include more covariates to decrease the variance of the error terms is unnecessary. Nevertheless, including additional covariates may be informative.

We estimate specifications that include the full set of four additional control variables, referred to as the extended models, in Table 3. To conserve space, for all control variables, we report only their estimated direct and indirect effects. Our main focus remains on the estimates for the peer-effect parameter ρ. All other details related to the estimation parallel the more parsimonious Table 2 models.

The coefficients in both panels of Table 3 largely echo the earlier findings. In particular, the coefficients on the peer-effect parameter ρ are positive and significant in all models, with magnitudes that are comparable to the corresponding earlier estimates. In the case of market leverage levels (column 1), the coefficients remain quite close to the 0.20 level, with a small decrease when turning to book leverage (column 2, estimates of 0.152 and 0.128 in the two panels). The role of the added control variables is largely consistent with prior research. As we detail in the Appendix (Table A.2), the estimates on the peer-effects parameter ρ are also substantively unchanged when we estimate parallel SDM models that allow for possible contextual effects.

The standard errors in the preceding analysis are derived using the MCMC approach which exploits numerical features of the likelihood function. A reasonable alternative is to use a simulation-based approach in which uninformative peer-weighting matrices are constructed and p-values are derived from the resulting distribution of peer-effect parameter estimates (ρ) under the null. We have undertaken this exercise using peer matrices $\tilde{W}$ that are randomly drawn (100 draws per model), but which stochastically resemble the true peer matrix W on several dimensions (details in the Appendix). The resulting peer-effect estimated coefficients are centered around 0, and in all

[12] Similar to LR, the relation between equity shocks and debt issuance is often anomalous.

Table 4
Robustness of peer effects findings to alternative modeling. The table presents estimates for models that are modifications of the parsimonious SAR models of peer effects in capital structure presented in Table 2. The first row of figures in each panel reports peer-effect coefficient estimates ρ for a model in which the industry and year fixed effects are replaced with firm and year fixed effects. The second row of figures in each panel reports peer-effect coefficient estimates ρ for a model in which the industry and year fixed effects are replaced with industry×year fixed effects. Standard errors calculated as in the prior tables using the MCMC approach and the numerical properties of the likelihood function are reported in brackets under each coefficient estimate. Standard errors calculated using a simulation approach in which uninformative peer-weighting matrices (details in Appendix) are used to derive the distribution of peer-effect parameter estimates (ρ) under the null are reported in parentheses under the MCMC standard errors. All other model, sample, and estimation details are as described in the note to Table 2.

	Market leverage	Book leverage	Equity issuance	Debt issuance	Δ Market leverage	Δ Book leverage
Panel A. Equity shock exogenous variable						
Peer-effect coefficient ρ	0.115***	0.057***	0.055***	0.043***	0.119***	0.040***
(Firm and year fixed effects)	[0.004]	[0.005]	[0.004]	[0.005]	[0.004]	[0.004]
	(0.007)	(0.008)	(0.006)	(0.005)	(0.005)	(0.006)
Peer-effect coefficient ρ	0.228***	0.172***	0.115***	0.038***	0.051***	0.010**
(Industry×year fixed effects)	[0.004]	[0.004]	[0.005]	[0.005]	[0.004]	[0.005]
	(0.008)	(0.008)	(0.006)	(0.005)	(0.005)	(0.006)
Panel B. Debt due exogenous variable						
Peer-effect coefficient ρ	0.156***	0.056***	0.081***	0.056***	0.146***	0.050***
(Firm and year fixed effects)	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Peer-effect coefficient ρ	0.233***	0.123***	0.140***	0.042***	0.082***	0.016***
(Industry×year fixed effects)	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
	(0.006)	(0.006)	(0.004)	(0.004)	(0.005)	(0.004)

Significance at * 10% level, ** 5% level, *** 1% level

cases the actual estimate reported in Table 2 exceeds the 99th percentile of the derived empirical distribution under the null.

Following LR, our default choice in the prior models is to include industry and year effects. As reasonable alternatives, we modify the baseline models to include firm and year effects, followed by industry × year effects. We report the resulting peer-effect coefficient (ρ) estimates using the two exogenous shifters of capital structure in the two panels of Table 4, with standard errors calculated using both the MLE MCMC approach (in brackets) and the simulation approach (in parentheses). As the figures reveal, the estimated peer-effect coefficients remain positive and highly significant in all models. In the key leverage level models (columns 1 and 2), the estimated magnitudes do drop moderately when including firm and year effects, while remaining substantively unchanged from the baseline estimates when the model includes industry × year effects.

4.4. Comparison to evidence from alternative estimation approaches

The evidence we present above indicates an economically meaningful structural peer-effect coefficient estimate for leverage levels of around 0.20.[13] As we discuss earlier, the LR evidence is consistent with the presence of significant causal peer effects in capital structure. However, the magnitudes of their reported 2SLS estimates do not have any clear structural economic interpretation, although they do suggest such an interpretation. In particular, they suggest that a one-unit change in peer average leverage results in a change in a firm's leverage in the 0.7 to 1.0 unit range.[14] This reported effect is at least 3.5 to 5 times larger than what is implied by our estimates and suggests that peer leverage is a dominant factor in causally determining a firm's leverage policy.

This startling difference in the suggested magnitude of peer effects raises the general issue of the sensitivity of coefficient estimates in peer-effects settings to various modeling/sampling choices. To investigate, we conduct a 2SLS that parallels the LR modeling of leverage levels. As we report in the first two columns of Table 5, the 2SLS coefficients we estimate on peer (unweighted) average leverage, reported in the first row, agree with LR in sign, but they are small and insignificant.[15] Since our 1996–2017 sample period only partially overlaps with LR (1965–2008), we also estimate 2SLS models for this earlier period. As we report in columns 3–4 of Table 5, the estimated peer-effects coefficients are somewhat different for this earlier period,

[13] Again, because of the feedback feature of peer effects, this 0.20 figure is the first step of a geometric process that will result in a slightly higher than 0.20 increase in leverage after an initial perturbation of the weighted peer average. In the case of a symmetric network, the total effect will be on the order of 0.20/(1-0.20)=0.25

[14] For market leverage levels, LR report a 0.727 coefficient in their Table 8. For book leverage levels, the adjusted coefficient they report in Table 7 implies a coefficient of slightly above 1.0

[15] As in LR, the equity shock variable is highly significant in all first-stage regressions.

Table 5
Peer-coefficient estimates from traditional 2SLS models. This table presents 2SLS estimates predicting firms' leverage levels as a function of peer average leverage and control variables. Peer average leverage is instrumented using the average equity shock of a firm's industry peers and the peer set is defined based on common 3-digit SIC codes. Coefficient estimates from the second stage are reported in the table with standard errors under the coefficient estimates. The peer firm control variables are simple averages of the set of same-industry peers, and all control variables are defined as in the earlier tables. For all variables except the peer firm dependent variable, the estimates are scaled by the variable's standard deviation. All models include industry and year fixed effects and standard errors under the coefficient estimates are clustered at the firm level. Modeling choices in Panel A are designed to parallel the treatment in Table 8 of Leary and Roberts (2014). The models in columns 1–2 of Panel A are for our main sample period of 1996–2017, and the models in columns 3–4 are estimated on a sample that is extended back to 1965 and truncated at 2008. In Panel B, we record in column 1 the estimate for the 2SLS peer dependent variable coefficient taken directly from the market leverage model of Leary and Roberts (2014), and in column 2 we report our corresponding estimate for the same dependent variable and sample period. The subsequent Panel B columns are for the same sample period and dependent variable, but in column 3 we drop all of the control variables, in column 4 we include only tangibility as a control variable, and in column 5 we include all control variables except tangibility. In Panel B the estimated coefficients on the control variables are omitted from the table for brevity.

Panel A. Predicting market and book leverage with full set of controls

	1996–2017		1965–2008	
	Market leverage	Book leverage	Market leverage	Book leverage
Peer firm				
Dependent variable	0.026	0.119	0.249***	0.029
	[0.108]	[0.207]	[0.086]	[0.105]
Own firm variables				
Equity shock	−0.010***	−0.006***	−0.026***	−0.011***
	[0.001]	[0.001]	[0.001]	[0.001]
Log(sales)	0.031***	0.045***	0.009***	0.010***
	[0.001]	[0.001]	[0.001]	[0.001]
Market-to-book	−0.068***	−0.005***	−0.058***	−0.015***
	[0.001]	[0.001]	[0.001]	[0.001]
Profit	−0.039***	−0.045***	−0.288***	−0.248***
	[0.001]	[0.001]	[0.011]	[0.012]
Tangibility	0.046***	0.051***	0.173***	0.201***
	[0.001]	[0.001]	[0.014]	[0.013]
Peer firm variables				
Log(sales)	0.004**	−0.003**	0.017***	0.024***
	[0.002]	[0.002]	[0.004]	[0.004]
Market-to-book	−0.013***	−0.001	0.039***	−0.038***
	[0.002]	[0.002]	[0.009]	[0.009]
Profit	−0.013***	−0.001	0.006***	0.005***
	[0.002]	[0.002]	[0.002]	[0.002]
Tangibility	0.011***	0.009***	0.002	0.002
	[0.002]	[0.002]	[0.005]	[0.003]

Panel B. Model permutations predicting market leverage

Model permutation	(1) LR	(2) All controls	(3) No controls	(4) Tangibility only	(5) No tangibility
Peer mkt. lev.	0.727***	0.249***	0.796***	0.882***	0.275***
	[0.151]	[0.086]	[0.123]	[0.088]	[0.110]

Significance at * 10% level, ** 5% level, *** 1% level

and the coefficient in the market leverage model (column 3) is now significant. However, even this coefficient (0.249) is substantially smaller than the corresponding LR figure.

This evidence suggests that 2SLS estimates of the coefficient on peer average leverage can be quite sensitive to small variations in sampling or modeling choices. This is perhaps not surprising, given that the coefficient on the peer average leverage is a nonlinear function of both the underlying peer-effect coefficient *and* other model coefficients. To further explore this issue, we estimate 2SLS market leverage models for the LR sample period using a variety of model permutations, some of which are reported in Panel B of Table 5. The first column of this panel lists the 0.727 estimate taken directly from LR, while the second column re-lists our corresponding estimate of 0.249. If we drop all of the (possibly endogenous) control variables, our peer-coefficient estimate jumps up sharply to 0.796 as reported in column 3, and now appears fairly close to the LR figure. If we add just the tangibility control variable, this figure jumps up further to 0.882, as indicated in column 4. However, if we include all control variables except tangibility, the resulting coefficient drops to a much smaller 0.275 as reported in column 5 of the panel. This wide variation in estimates across seemingly small modifications in modeling choices certainly raises concerns about the 2SLS approach.

To further explore, we consider 3 different winsorization treatments (1%, 2%, 3%), a full set of 16 different in-

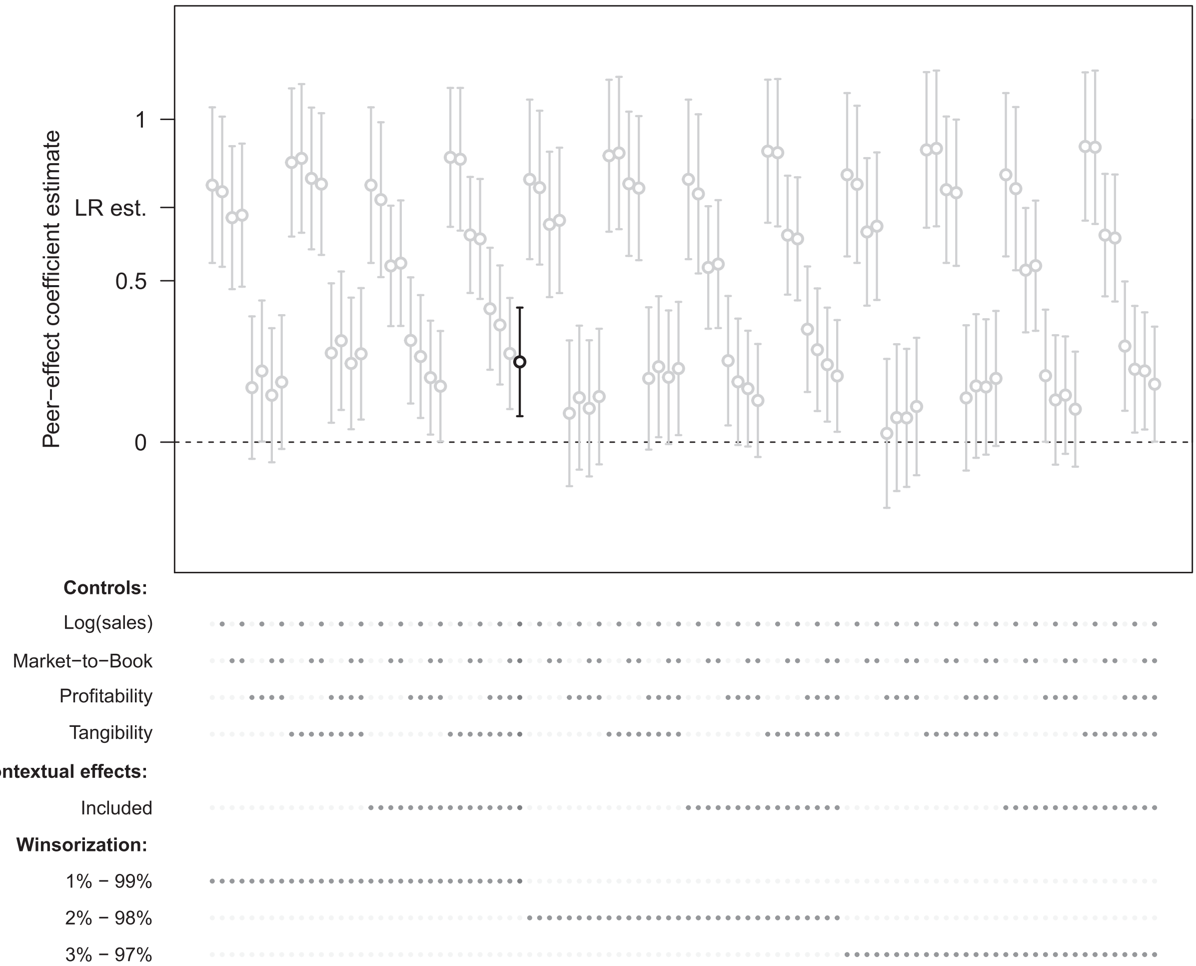

Fig. 3. Frequency distribution of 2SLS peer-coefficient estimates. This figure presents the distribution of 2SLS peer dependent-variable coefficient estimates predicting market leverage levels from 1965 to 2008 as a function of peer average leverage and control variables, with peer average leverage instrumented using equity shocks. The distribution is based on 96 model permutations outlined in the text and the precise included elements for each specification corresponding to each estimate is indicated by the legend below the figure. For each permutation, the circle corresponds to the point estimate and bars represent a 95%-confidence interval. The highlighted specification indicates the estimate for the baseline model in column 2 of Table 5. For comparison purposes, the corresponding LR estimate of 0.727 is noted on the y-axis.

clusion/exclusion choices for the 4 control variables, and the inclusion/omission of corresponding contextual effects. This yields $3 \times 16 \times 2 = 96$ model permutations. For each permutation, we estimate a 2SLS model explaining market leverage and report in Fig. 3 the resulting peer-effect ρ coefficient estimates and associated confidence intervals.[16] As is clear from the figure, these relatively small variations in modeling choices result in a set of estimates that vary widely, from close to 0 to more than 0.90.[17] Clearly, the dispersion in these figures suggests that 2SLS estimates of purported peer-effect coefficients are quite unstable, in addition to being theoretically ambiguous indicators of the underlying quantity of interest.

4.5. Evidence on the role of the peer-network characterization

In addition to the novel estimation approach, the second distinguishing feature of our study is the use of product similarity scores to characterize the peer network. This approach is quite different from the traditional reliance on SIC codes, a strategy that necessarily restricts the peer set to have a transitive structure and does not readily allow peers to be weighted differentially. To investigate the role that the peer network characterization has for inferences, we consider a set of spatial models using a range of characterizations that effectively span the traditional SIC approach and our approach.

[16] This approach of considering the entire set of coefficient estimates derived from a wide range of defensible specification choices to achieve robust inferences follows the spirit of the recommendation by Simonsohn et al. (2020).

[17] The evidence on substantial variation in peer-effects 2SLS coefficient estimates also holds if we consider book leverage levels and/or our more recent sample period, although the maximum estimates in these cases are smaller than what is illustrated in Fig. 3.

Table 6
Peer-coefficient estimates for alternative peer network characterizations. This table presents coefficient estimates for spatial models for the 1996 to 2017 period predicting levels of market and book leverage as a function of peer leverage choices, equity shocks, and the full set of four additional control variables and peer values for these control variables (i.e., contextual effects). For brevity, only the estimates for the peer-coefficients are tabulated. Standard errors derived from the likelihood function are reported in brackets under each estimate and all models are estimated as MLE models. The peer-weighting matrix W varies across models as noted in the final row of the table. Columns 1 and 2 are for our default W matrix based on the level of similarity scores for the ten closest peers. Columns 3 to 6 weight all firms in the same 3-digit SIC code equally. In columns 3 and 4 the nonpeer coefficients are mechanically set equal to their values in the corresponding column 1 and 2 estimation, while in columns 5 and 6 they are estimated. Columns 7 and 8 choose the peer set based on 3-digit SIC codes but then weight the members in that set proportional to their product similarity scores. Columns 9 and 10 weight all peers equally within the 10 closest peer set based on product similarity scores. Columns 11 and 12 include all peers with nonzero levels of similarity and weigh each firm in this set proportional to the similarity level. All other model, sample, and estimation details are as described in the note to Table 2.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.
Peer-effect coefficient ρ	0.220***	0.179***	0.171***	0.132***	0.160***	0.124***	0.165***	0.135***	0.223***	0.180***	0.231***	0.197***
	[0.005]	[0.005]	[0.004]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Log marg. likelihood	−63,242.16	−65,599.95	−66,675.96	−68,979.34	−66,687.12	−68,996.96	−67,244.82	−69,555.63	-63,404.78	-65,759.91	-63,944.93	-66,256.99
Peer network	Default		SIC, equal weighted (fixed coefficients)		SIC, equal weighted (estimated coefficients)		SIC, similarity weighted		Most similar 10 peers equal weighted		Full peer set similarity weighted	

Significance at * 10% level, ** 5% level, *** 1% level

As a baseline, we report in models 1–2 of Table 6 estimated peer-effect coefficients from an initial model that uses our default peer-network matrix, referred to as $W_{default}$, with a slight adjustment to the explanatory variable structure to parallel the LR models.[18] To ease comparisons, we restrict the sample to a set of common observations that are available for all models. As expected, the peer-effect coefficients ρ in these models are quite close to the earlier estimates, hovering around 0.20.[19]

We also report in the table log-marginal likelihood (LML) values to assess model fit. These values provide a natural way to conduct Bayesian model comparisons. Specifically, for two models M_1 and M_2, the ratio of marginal likelihoods ML_1/ML_2, known as the Bayes factor, represents the ratio of the probability that model 1 is the correct model relative to the probability that model 2 is the correct model. Therefore, larger (i.e., less negative) figures for the log-marginal likelihood values provide evidence of better model fit. Since log-marginal likelihood values are exponentiated in calculating the Bayes factor, relative model fit will be highly sensitive to these values, as the Bayes factor will equal $e^{LML_1 - LML_2}$. Practically speaking, this means that in all cases we tabulate below, the model with the less negative LML value will be selected as the preferred model compared to a model with a more negative LML value with a very high (relative) probability.[20]

We proceed by adopting the LR approach to modeling peer interaction by constructing a peer matrix $W_{sictransitive}$ that weights all firms in the same SIC 3-digit industry equally (i.e., weights of 1/N for the N other firms in the same SIC industry-year). To estimate the peer coefficient ρ using this peer network, we fix all other model coefficients at their estimated values from the corresponding baseline model. The identification problem is thus sidestepped by relying on the strong assumption that the other model coefficients are known. As we report for models 3–4 of Table 6, estimated peer effects assuming this peer network structure are roughly three-quarters of what we estimate in the initial baseline models 1–2.

To explore further, in models 5–6 we continue using the $W_{sictransitive}$ peer network, but we allow all parameters to be estimated. This model overcomes the identification problem discussed earlier for transitive networks by exploiting variation in the number of firms in each industry. Models of this type are often considered only weakly identified, with a debate as to whether the resulting estimates are informative (see, e.g., Lee, 2007; Bramoullé et al., 2009). The peer-effect coefficients reported for these models are again smaller than what we find when using the

[18] In these models, we follow the LR hybrid explanatory variable structure in that we do not allow equity shock contextual effects (following the SAR model), but we do allow for contextual effects for the other explanatory variables (following the SDM model).

[19] All of the evidence in Table 6 is substantively unaltered if we use a strict SAR or SDM approach and/or if we exploit the *debtdue* variable as the exogenous shifter of capital structure.

[20] We do not report Bayes factors in the table, as they are in all cases very large numbers for the choice of a less negative LML model compared to a more negative LML model, and approximately zero for the reverse comparison.

Table 7

Noise in the peer-network characterization. This table presents estimates for the peer-effect coefficient for SAR spatial models for the 1996 to 2017 period predicting simulated levels of leverage as a function of peer leverage choices, an exogenous variable, and industry and year fixed effects. In Panel A, the exogenous variable is a firm's lagged idiosyncratic equity shocks. In Panel B, the exogenous variable is the fraction of the firm's long-term debt that is due in the coming year. The peer-weighting matrix W varies across models as noted in the peer network row. The networks used in columns 1–5 are described in Table 6. In columns 6–8, we combine the default peer-weighting matrix with a pure noise matrix constructed as described in the text, with the relative weight placed on the noise matrix indicated in the peer network row. For each model, 100 simulated data sets for the dependent variable are created by fixing coefficient estimates at their Table 2, column 1 values (and thus ρ is set to 0.250) and drawing an error term for each observation from a normal distribution with the variance equal to the variance of the residuals from the Table 2 model. We estimate the peer-effect coefficient ρ for all 100 simulated samples for each peer network characterization and report the median coefficient estimate and associated standard error in the table. The sample is restricted to 38,524 observations from 1996 to 2017 with data available for all of the different peer network characterizations. All other estimation details are described in the earlier tables.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Equity shock exogenous variable								
Peer-effect coefficient ρ	0.248*** [0.007]	0.103*** [0.007]	0.096*** [0.007]	0.255*** [0.007]	0.262*** [0.010]	0.277*** [0.010]	0.148*** [0.011]	0.048*** [0.009]
Log marg. likelihood	−67,054.65	−67,260.11	−67,734.12	−67,136.53	−67,105.73	−67,230.40	−67,698.36	-67,790.41
Peer network	Default	SIC equal weighted	SIC similarity weighted	Most similar 10 peers equal weighted	Full peer set similarity weighted	50% noise	75% noise	90% noise
Panel B. Debt due exogenous variable								
Peer-effect coefficient ρ	0.256*** [0.006]	0.103*** [0.008]	0.099*** [0.007]	0.261*** [0.007]	0.268*** [0.011]	0.272*** [0.010]	0.164*** [0.011]	0.073*** [0.010]
Log marg. likelihood	−52,010.16	−52,280.60	−52,632.11	−52,029.29	−52,155.31	−52,233.17	−52,574.87	-52,615.70
Peer network	Default	SIC equal weighted	SIC similarity weighted	Most similar 10 peers equal weighted	Full peer set similarity weighted	50% noise	75% noise	90% noise

Significance at * 10% level, ** 5% level, *** 1% level

Table 8

Peer-effect coefficient estimates by subsample. This table presents subsample estimates of the peer-effect coefficients ρ from the parsimonious SAR models of Table 2 in which corporate leverage levels are a function of peer leverage and an exogenous explanatory variable. The first (last) two columns of estimates use the equity shock (*debtdue*) variable as the exogenous catalyst of capital structure changes. Panel A splits the sample by whether a firm is in an industry with above/below median level of concentration as measured by a start of year sales-based HHI index. Panel B splits the sample by whether a firm operates in an environment with above/below median level of fluidity as measured by the start-of-year Hoberg and Phillips (2016) fluidity score. Panel C splits the sample by whether the firm operates in an industry with an above/below level of private firms according to U.S. Census Bureau data. Panel D compares subsamples based on whether the firm operates in an NAICS 3-digit industry with above/below median levels of leverage as of the start of the observation year. Panel E considers only firms with sales at above the sample 30th percentile level. The sample period is 1996–2017 and all other model, sample, and estimation details are as described in the note to Table 2.

	Equity shock		Debt due	
	Mkt. lev.	Book lev.	Mkt. lev.	Book lev.
Panel A. Industry concentration				
High concentration	0.208***	0.173***	0.207***	0.128***
	[0.006]	[0.006]	[0.005]	[0.006]
Low concentration	0.231***	0.139***	0.246***	0.111***
	[0.006]	[0.006]	[0.006]	[0.006]
Panel B. Industry fluidity				
High fluidity	0.197***	0.119***	0.190***	0.087***
	[0.008]	[0.008]	[0.007]	[0.007]
Low fluidity	0.169***	0.120***	0.161***	0.083***
	[0.007]	[0.006]	[0.006]	[0.006]
Panel C. Proportion of public firms				
Low private	0.215***	0.135***	0.213***	0.085***
	[0.006]	[0.005]	[0.005]	[0.005]
High private	0.213***	0.167***	0.185***	0.087***
	[0.012]	[0.012]	[0.010]	[0.011]
Panel D. Industry leverage level				
High leverage	0.211***	0.171***	0.210***	0.123***
	[0.008]	[0.009]	[0.007]	[0.008]
Low leverage	0.159***	0.088***	0.151***	0.053***
	[0.006]	[0.006]	[0.005]	[0.005]
Panel E. Excluding small caps				
Restricted	0.259***	0.211***	0.264***	0.171***
	[0.005]	[0.005]	[0.004]	[0.004]

Significance at * 10% level, ** 5% level, *** 1% level

default peer network structure, and the model fit as indicated by log-marginal likelihood values is substantially poorer compared to the baseline model. This evidence suggests that our default modeling of peer interactions based on product similarity scores leads to substantially more informative estimates than a simple SIC code approach.

It is unclear whether the better model fit using the default peer network characterization primarily reflects how the peers are identified, or how they are weighted. To provide some evidence, we consider a peer matrix in which the members of the peer network are selected based on common SIC codes, but the weighting within this set is assumed to be proportional to product similarity scores.[21] As we report in models 7–8 of Table 6, estimated coefficients when assuming this peer network structure are noticeably smaller than in our default models, and the fit is substantially worse. This strongly suggests that important information content is gained when the members of the peer network are selected based on product similarity scores rather than SIC codes.

To investigate the role of weights, we next select the peer set using our default approach, but create a peer-network matrix that weights all of the identified peers equally (rather than proportional to similarity scores). The estimated peer-effect coefficients for this choice, as reported in models 9–10 of Table 6, jump up to a similar level to our baseline model, with a fit that is much better than the SIC-based models, but somewhat worse than our default approach. This evidence suggests that the more important element in our peer network characterization is in the selection of peers, rather than in the precise weights.

Rather than simply restricting attention to the 10 closest peers, we also consider extending the peer network to include all firms with nonzero levels of product similarity. As we report in models 11–12 of Table 6, when we employ a matrix with this peer network characterization, the magnitude of the estimated peer-effect coefficients ρ are similar to our baseline model estimates, but again with a somewhat poorer model fit. It appears that inclusion of distant peers does not lead to an improvement that would merit their inclusion.[22]

Summarizing, our baseline models indicate a significant peer role in leverage decisions with an initial effect on the order of 0.20 that is then slightly magnified as the firm and peer network re-equilibrate. Our default representation of peer interactions based on cardinal measures of product similarity scores restricted to the ten closest peers appears to fit the data the best. However, the magnitudes of estimated peer-effect coefficients are insensitive to reasonable changes in the peer-weighting matrix, provided that product similarity is used to define the peer network.

4.6. Measurement error in the peer network

The preceding evidence that estimates are not highly sensitive to precise similarity scores is comforting, as these scores surely measure the degree of peer interaction with some error. To further investigate the potential role of measurement error in the peer network characterization, we experiment with using simulated data in which the actual peer matrix is known. By adding noise to this matrix, we can garner insights on the sensitivity of model estimates to errors in the network characterization.

To simulate leverage choices, we select our default peer-weighting matrix $W_{default}$ and require leverage to be

[21] These scores are obtained from the full Hoberg and Phillips' TNIC database. Note that proportional weighting by similarity scores is the approach incorporated into our default peer-weighting matrix.

[22] If we use the top-20 or top-50 peers rather than the top-10, the resulting coefficients and measures of fit fall somewhere between the top-10 default model and this full-set-of-peers alternative model. Note that Hoberg and Phillips (2016) assign similarity scores of 0 for all firm pairs in which a raw measure of similarity falls below a minimum threshold.

governed by the SAR model, with equity shocks as the sole explanatory variable.[23] We mechanically fix ρ to equal 0.250 (to match the model in column 1 of Panel A of Table 2) and set the coefficient on the equity shock variable to its estimated value in this same model. We then draw the error term for each observation from a normal distribution with the variance set equal to the estimated variance of the residuals in the tabulated Table 2 model. Using the resulting simulated sample, we estimate a peer-effect coefficient. We repeat this simulation 100 times and report the median estimates in Panel A of Table 7.

Starting with the correct peer-network matrix (i.e., $W_{default}$), the peer coefficient estimate using the simulated sample is, as reported in model 1 of Panel A of Table 7, precisely estimated and quite close to the true value (0.248 vs. 0.250). Next, we estimate the model on the simulated data using the four alternative peer matrices discussed above (two utilizing SIC code information and two with alternative weights placed on product similarity scores relative to our default choice). As the figures in the table illustrate, the peer matrices that rely on SIC codes result in estimated coefficients that are less than half of their true values (see models 2 and 3), and with a marked decrease in model fit. The models that rely exclusively on product similarity scores yield coefficients that are quite close to the true value (0.255 and 0.262 as reported in models 4 and 5), and the decrease in model fit is only moderate.

Since the level of noise in these models is of unknown character, we next experiment with injecting pure mechanical noise into the peer-weighting matrix. To do this, we create a completely uninformative noise matrix by randomly assigning peers to firms.[24] We then create a noisy version of the true peer-network matrix by taking the simple average of the correct peer-network matrix and the pure noise matrix. As we report in model 6 of Panel A of Table 7, the estimated peer coefficient on the simulated sample using this half-truth/half-noise peer network remains fairly close to the true value (0.277 versus 0.250), but the model fit is, as expected, poorer than when we use the correct network matrix.[25] If we turn up the noise to the 25%/75% level, the coefficient starts to show significant downward bias, with an estimate that is approximately 60% of the true value, as revealed in model 7 of Panel A of Table 7. At a 10%/90% noise ratio, the coefficient is badly downward biased (see model 8), and it asymptotically approaches 0 as noise increases above this level. The evidence in Panel B of the table, which instead uses long-term debt as the exogenous explanatory variable, tells the same story.

Collectively, the evidence in Tables 6 and 7 suggest that the magnitudes of estimated peer-effect coefficients using our framework are not highly sensitive to a reasonable degree of noise in the matrix characterizing peer interactions. However, there is a significant difference between using SIC codes and product similarity scores as the starting point for characterizing peer interactions. The evidence we present suggests that the latter choice will generally lead to more economically informative estimates.

4.7. Variation across subsamples

The findings above provide compelling evidence for the presence of substantive peer effects in capital structure decisions. To help understand the economic mechanism underlying our sample-wide findings, we compare estimates derived from subsamples sorted by characteristics that may influence the magnitude of peer effects under certain scenarios. Since strategic effects may be more pronounced in highly concentrated market settings, we first separate the sample by a measure of industry concentration. Specifically, at the start of each year, we rank industries by sales-based HHI index scores, calculated at the Hoberg and Phillips FIC industry level, and split the sample by the annual cohort median.

As we report in Panel A of Table 8, there are no compelling differences in peer-effect coefficients ρ estimated for more versus less concentrated industries. The book leverage models suggest a slightly higher peer effect in more concentrated industries, while the market leverage models suggest the opposite. Given the limitations of the HHI index to capture the nuanced nature of variation in strategic interactions, this lack of a clear difference across subsamples is not entirely surprising.

As an alternative to measures of concentration, we investigate the possibility that peer effects are larger in more dynamic economic environments in which the risks to ignoring peer capital structure could be particularly large. We do this by separating the sample based on whether a firm operates in a more versus less fluid product-market environment, as measured by the Hoberg and Phillips (2016) fluidity score. This measure is often associated with the level of competitive threats and rate of product-market change. As we report in Panel B of Table 8, we detect some modest evidence of elevated peer effects for firms with above median fluidity compared to the below median group. In the case of market leverage, estimated peer effects for the former group are approximately 1.15 times the corresponding estimates for the latter group. However, no meaningful differences are evident from the corresponding book leverage models.

A limitation of the preceding analysis is that it does not account for variation that may arise from the presence of private firms. To explore, we examine subsamples based on whether a firm's NAICS 3-digit industry includes an above or below median percentage of private firms according to U.S. Census Bureau data. As we report in Panel C of Table 8, differences across these subsamples are small in magnitude and switch in sign across various models, casting doubt on the hypothesis that private firms play an important role in our findings. Since financing decisions of private firms are not publicly disclosed and most private firms are relatively small, this may reflect the inability of

[23] The character of our findings are substantively unchanged if we add the full set of control variables or allow for an SDM model structure.

[24] Each true connection is replaced with a connection of the same magnitude between two randomly selected firms.

[25] The usual intuition that increased noise results in a downward bias in coefficient estimates does not universally apply in spatial models owing to the nonlinearity in the relationship between reduced form coefficients and structural parameters.

Table 9
Peer effects in other corporate finance settings. This table reports estimates from SAR (Panel A) and SDM (Panel B) models predicting the indicated firm-level dependent variable as a function of peer dependent-variable choices and a set of explanatory variables that vary depending on the dependent variable as detailed in the text and the Appendix. All models are estimated using MLE implemented by an MCMC approach. The sample period is 1996–2017 and all models include industry and year fixed effects. The peer-weighting matrix is the default matrix W described in the text based on product similarity scores for a firm's closest 10 peers. The peer-effect coefficient ρ estimates are reported in the table with standard errors based on numerical features of the likelihood function in brackets under the reported coefficients. All other coefficient estimates are left untabulated for brevity. The figures on changes in nonpeer coefficients present information on the spatial model coefficient estimates on other explanatory variables relative to their estimated values from corresponding OLS models that do not allow for peer effects. For each corresponding pair of coefficient estimates, we calculate the difference in the magnitude (absolute value) of the larger magnitude estimate minus the smaller, all divided by the larger. The reported figures on median changes are the median of this percent difference across all control variables in any given model.

	Capex	R&D	Asset growth	Cash	Div. yield
			Panel A. SAR		
Peer-effect coefficient ρ	0.282***	0.224***	0.121***	0.264***	0.128***
	[0.004]	[0.003]	[0.004]	[0.004]	[0.004]
Median percent change in nonpeer coefficients	45.1%	40.1%	111.0%	14.7%	89.2%
			Panel B. SDM		
Peer-effect coefficient ρ	0.294***	0.286***	0.141***	0.242***	0.129***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Median percent change in nonpeer coefficients	56.0%	35.0%	101.5%	22.0%	91.2%

Significance at * 10% level, ** 5% level, *** 1% level

private-firm leverage choices to serve a meaningful strategic commitment role in an industry.

If an industry generally displays high leverage levels, one might suspect that industry participants will have a wider range of feasible leverage adjustments if they seek to respond to the actions of peers. Moreover, for firms in these industries, it seems likely that the firm's objective function near the optimum will tend to be relatively flat with respect to variations in their leverage choice. These considerations could result in relatively larger leverage adjustments for firms in response to a peer's choice that perturbs the parameters of the firm's optimization problem. As we report in Panel D of Table 8, the evidence strongly supports these suspicions. In particular, estimated peer-effect coefficients in the market (book) leverage models are approximately 25% (100%) larger for firms in NAICS 3-digit industries with above median leverage compared to firms in below median industries.

If small private firms do not treat leverage as a strategic choice variable in the context of product-market interactions, we may expect similar behavior for small public firms. To investigate, in Panel E of Table 8, we eliminate all sample observations for firms that have sales below the 30th percentile in a given year. As this panel illustrates, this restriction uniformly raises the magnitude of estimated peer effects by a modest amount, with the ρ coefficients averaging 0.225% or 22.5% across models, as compared to 0.20% or 20% in the corresponding models for the entire sample as reported earlier.

5. Peer effects in other finance contexts

As we discuss earlier, the key innovation in our study is the use of heterogeneity in peer connections and spatial econometrics techniques to identify causal structural parameters of interest (i.e., the peer-effect coefficient ρ). Importantly, these parameters parsimoniously characterize equilibrium firm behavior in response to any relevant exogenous variation that reverberates through the peer network. Given our capital structure evidence, it seems likely that the methodological perspective we apply could also be applied productively in other corporate finance contexts in which peer effects are suspected to be present.

While a full-scale analysis is beyond the scope of this paper, we briefly consider this issue here. The challenge in this exercise, compared to the capital structure investigation above, is that we do not have the benefit of a widely available off-the-shelf exogenous variable for these other settings to convincingly rule out various omitted variable and common shock channels. Nonetheless, an exploratory analysis may still be informative.

To conduct this investigation, we borrow the models related to investment decisions and financing identified by Grieser and Hadlock (2019) in their survey of common corporate finance specifications. These include canonical models of capital spending and Research and Development (R&D), augmented with a model of total asset growth. On the financing side, we model cash holdings and dividends as a function of the explanatory variables in the Grieser and Hadlock (2019) study. Since the economic content of these variables is not the focus, we relegate additional details to the Appendix.

We modify these off-the-shelf OLS models ($y = X\beta + \epsilon$) by adding peer effects to form the corresponding SAR and SDM models ($y = \rho Wy + X\beta + \epsilon$ and $y = \rho Wy + X\beta + WX\theta + \epsilon$). Using the same econometric approach as outlined in our capital structure analysis, we report in Table 9 the resulting estimates for the peer effects coefficients (ρ) (SAR estimates in Panel A, SDM estimates in Panel B) in each model. As the table reveals, all of the estimated ρ coefficients are positive, substantial in magnitude, and highly significant. This preliminary evidence is highly consistent with the presence of substantial peer effects in many key financial variables, with the aforementioned caveat concerning the exogeneity assumption for the covariates.

To compare OLS estimates to the derived effects from the peer-effects models, for each covariate, we calculate the percentage difference between the coefficient for the covariate in the OLS and corresponding spatial models, calculated as the difference in the magnitude (absolute value) of the larger magnitude estimate minus the smaller, all divided by the larger. We report the median of these ratios across the covariates in each estimated model in the second row of each panel in Table 9. In many cases, these differences are substantial (i.e., greater than 50%), suggesting frequent disagreements between OLS estimates and corresponding estimates from models that allow for peer effects. Given the possible presence of peer effects in many corporate finance settings, these differences suggest that, even if a researcher is uninterested in the role of peers, it may still be necessary to account for their presence in the modeling to uncover consistent estimates for the main (nonpeer-related) behavior of interest.

While we focus above on corporate finance applications, a growing literature suggests that peers likely play a prominent role in many individual investment decisions, ranging from stock selection to housing choices (e.g., Heimer, 2016; Bailey et al., 2018). In these settings, peer networks are typically intransitive in structure and straightforward to identify from social network data. Thus, it seems likely that the methods we exploit above could be used productively in many of these settings, with the potential to yield novel insights into the aggregate effect of individual-level shocks as they reverberate through a given network.[26]

6. Conclusion

In this study, we considerably clarify the picture on the role of peers in firms' capital structure choices. We identify peer networks and measures of the strength of firm interactions using the methodology developed by Hoberg and Phillips (2010, 2016). By exploiting the heterogeneous and intransitive nature of these networks, along with variation in an exogenous driver of capital structure popularized by Leary and Roberts (2014), we are able to overcome important unresolved identification challenges in prior work. Our estimation approach exploits recent developments in spatial econometrics to naturally extend models of geographic interactions to the case of product-market interactions.

We detect evidence of strong positive peer effects in capital structure decisions, with estimates indicating that the initial sensitivity of a firm's leverage choice to a similarity-weighted measure of peer average leverage is on the order of 0.20. These estimates indicate a substantive, but moderate, level of strategic complementarity in capital structure decisions. Our modeling allows peer effects to vary by a firm's location in the product-market network, with more connected firms having a relatively larger aggregate effect on their peers. Our findings appear robust to a variety of modeling alterations, including identifying peer effects using two distinct exogenous drivers of capital structure variation. Key elements of our empirical design strongly suggest that the behavior we detect reflects strategic considerations, and variation in the magnitude of estimated peer effects across different product-market environments helps strengthen this interpretation.

In addition to making a significant step forward in understanding peer effects in capital structure, we briefly explore the usefulness of our modeling approach for other questions in finance in which substantive interfirm peer interactions are suspected. Here, we detect the presence of significant peer effects for several other important firm policies. Moreover, the estimated role of other covariates in predicting outcome variables is often substantively affected by accommodating peer influences in the estimation framework. While this preliminary evidence should be interpreted with caution, as the case for exogeneity of the explanatory variables is less well established in these contexts, taken as a whole, the evidence strongly suggests that peer effects must be accounted for to uncover an accurate picture of firm financial behavior. The methodology we present provides a general framework for doing so.

Appendix A. Empirical estimation and interpretation

A1. Model estimation

A1.1. Transforming the model

We present the details for the SDM model, as the SAR model is subsumed by this model by adding the restriction of $\theta = 0$. For ease of exposition, we omit model intercepts as they can be incorporated into the explanatory variable matrix X as constants. After discussing the basic framework, we explicitly address different fixed-effects intercept structures given the panel-data features of the sample. Writing the SDM model in matrix form, we have

$$Y = \rho WY + \beta X + WX\theta + \epsilon. \tag{1}$$

Solving for reduced form equations in terms of Y, we can express the set of equations in (1) as:

$$Y = (I - \rho W)^{-1}(\beta X + WX\theta + \epsilon) \tag{2}$$

While the reduced form equations solve the simultaneity problem and allow us to identify the structural

[26] An added challenge in some of these contexts is that the peer network is often endogenously selected (see, for example, Cookson et al., 2021) and thus the empirical design would require the use of variation that is unrelated to this selection.

parameters, analytical solutions for the inverse are intractable for large networks. However, maximum likelihood estimation (MLE) implemented with an iterative Markov Chain Monte Carlo (MCMC) procedure can be utilized in a relatively straightforward manner. To implement this approach, we define $S=(I_N-\rho W)^{-1}$, $Z=[X\ WX]$, and $\delta=[\beta\ \theta]^T$. Assuming initially that ρ is known and equal to ρ^*, Eq. (2) can be rearranged as :

$$y_t-\rho^* W y_t = Z_t\delta+\epsilon_t. \tag{3}$$

Since $y_t-\rho^* W y_t$ is observable, Eq. (3) can be estimated using OLS to yield:

$$\hat{\delta}=(Z_t' Z_t)^{-1} Z_t'(I-\rho^* W)y_t, \tag{4}$$

with an estimated variance of the noise term σ^2 equal to $\hat{\sigma}^2=\frac{1}{N}\hat{\epsilon}_t'\hat{\epsilon}_t$, where $\hat{\epsilon}_t=(I-\rho^* W)y_t-Z_t\hat{\delta}$. The full (log) likelihood can therefore be concentrated with respect to the parameters β, θ, and σ^2. Maximizing the likelihood function reduces to a univariate nonlinear-optimization problem in the parameter ρ. Using the notation $\epsilon(\rho)$ to indicate that the estimate of vector ϵ depends on the value of ρ, the log-marginal likelihood can be expressed as:

$$ln[L(\rho)]=\kappa+ln(|I-\rho W|)-\frac{N}{2}[\epsilon(\rho)'\epsilon(\rho)], \tag{5}$$

where κ is a constant that does not vary with ρ. Eq. (5) can be maximized directly to obtain an MLE estimate of the parameter ρ. The actual maximization is implemented via the MCMC procedure detailed below.

A1.2. Adjusting for panel data

For simplicity, we have motivated our analysis above in a cross-sectional framework, but panel-data extensions are relatively straightforward. We denote a vector of fixed effects in time-period t by υ_t where υ_t is an $n_t\times 1$ vector and n_t is the number of firms in period t. Allowing a scalar time-specific fixed effect α_t leads to the following equation

$$y_t=\rho W_t y_t+X_t\beta+W_t X_t\theta+\upsilon_t+\alpha_t\iota_{n_t}+\varepsilon_t,\ \ t=1,\dots,T. \tag{6}$$

The total number of sample observations can be written as $N=\sum_{t=1}^T n_t$. The exogenous weight matrices W_t are of dimension $n_t\times n_t$ and are row-normalized to have row-sums of unity and zero elements on the diagonal. A nonzero (i,j) element of the matrix W_t indicates that firms i and j are related. The $K\times 1$ vector β of parameters is associated with the explanatory variables in the $n_t\times K$ matrix X_t. Given our construction of the matrices W_t, the matrix-product $W_t X_t$ produces a linear combination of peer-firms' explanatory variables, with the associated coefficient vector θ.

The disturbances are assumed to be independently and identically distributed with a scalar variance σ^2 that is constant across time periods. Restricting the parameters to be constant across time periods results in a model in which only the explanatory variables X_t and $W_t X_t$ at time t impact contemporaneous values of y_t, and consequently the data across periods can be jointly assembled to produce parameter estimates.

A matrix statement of this model is shown in Eq. (7) below, with accompanying descriptions of the underlying matrices and the resulting reduced form model listed in Eq. (8).

$$Y=\rho WY+X\beta+WX\theta+\Upsilon+E, \tag{7}$$

$$Y=\begin{pmatrix}y_1\\ \vdots\\ y_T\end{pmatrix},\ X=\begin{pmatrix}X_1\\ \vdots\\ X_T\end{pmatrix},\ E=\begin{pmatrix}\varepsilon_1\\ \vdots\\ \varepsilon_T\end{pmatrix}$$

$$\Upsilon=\begin{pmatrix}\upsilon_1\\ \vdots\\ \upsilon_T\end{pmatrix}+\begin{pmatrix}\alpha_1\iota_{n_1}\\ \vdots\\ \alpha_T\iota_{n_T}\end{pmatrix}$$

$$W=\begin{pmatrix}W_1 & 0_{n_1,n_2} & \dots & 0_{n_1,n_T}\\ 0_{n_2,n_1} & W_2 & \dots & 0_{n_2,n_T}\\ \vdots & & \ddots & \\ 0_{n_T,n_1} & \dots & & W_T\end{pmatrix}$$

$$Y=(I_N-\rho W)^{-1}(X\beta+WX\theta+\Upsilon+E) \tag{8}$$

We note that the $N\times N$ matrix inverse $(I_N-\rho W)^{-1}$ will reflect the block-diagonal nature of W and thus explanatory variables in each period t only impact contemporaneous own-firm and other-firm outcomes. The matrix inverse can be expressed as an infinite series as shown in Eq. (9).

$$(I_N-\rho W)^{-1}=I_N+\rho W+\rho^2W^2+\rho^3W^3+\dots \tag{9}$$

$$W^g=\begin{pmatrix}W_1^g & 0_{n_1,n_2} & \dots & 0_{n_1,n_T}\\ 0_{n_2,n_1} & W_2^g & \dots & 0_{n_2,n_T}\\ \vdots & & \ddots & \\ 0_{n_T,n_1} & \dots & & W_T^g\end{pmatrix},$$
$$g=1,\dots,\infty.$$

A1.3. Adjusting for fixed effects

Our approach for adjusting for fixed effects follows the familiar strategy of appropriately demeaning the dependent and explanatory variables, although the demeaning procedure is slightly more nuanced than in a traditional setting. In what follows, it is useful to rewrite Eq. (7) as:

$$(I_N-\rho W)Y=X\beta+WX\theta+\Upsilon+E. \tag{10}$$

We first consider the following two models:

$$Y=X\beta_1+WX\theta_1+\Upsilon_1+E_1,$$

and

$$WY=X\beta_2+WX\theta_2+\Upsilon_2+E_2.$$

These two equations represent linear models and therefore standard arguments that demeaning results in an equivalent model will apply. Assuming for the moment that only time fixed effects are present, i.e. $\upsilon_t=0$ for all t, cross-sectional demeaning results in an equivalent model. Formally, define $\ddot{Y}=Y-\overline{Y}$, $\ddot{X}=X-\overline{X}$, $\ddot{WX}=WX-\overline{WX}$, $\ddot{E}_1=E_1-\overline{E}_1$, and $\ddot{E}_2=E_2-\overline{E}_2$ with $\overline{Y},\overline{X}$, $\overline{WX}$, $\overline{E_1}$ and $\overline{E}_2$ representing cross-sectional averages. The demeaned equations will be given by

$$\ddot{Y}=\ddot{X}\beta_1+W\ddot{X}\theta_1+\ddot{E}_1,$$

and

$$\ddot{WY}=\ddot{X}\beta_2+W\ddot{X}\theta_2+\ddot{E}_2.$$

If we assume ρ is known, we can define $\beta = \beta_1 - \rho\beta_2$ and $\theta = \theta_1 - \rho\theta_2$. Taking the difference between the last two equations and using substitution yields:

$$(I_N - \rho W)\ddot{Y} = \ddot{X}\beta + W\ddot{X}\theta + \ddot{E}. \tag{11}$$

It is immediately evident that models (10) and (11) are equivalent for estimating the parameters β, θ, and σ conditional on a fixed ρ. However, since our estimation procedure (outlined below) iteratively fixes ρ to derive these other coefficient estimates, followed by a search for the likelihood optimizing ρ parameter, estimation of the demeaned model remains equivalent to the original model. For ease of exposition, and with a slight abuse of notation, we will refer to the demeaned variables as Y and X (instead of $\ddot{Y}$ and $\ddot{X}$) for the remainder of the paper. All alternative fixed-effects structures considered in the paper are removed via an analogous demeaning procedure.

A1.4. Implementing the estimation via MCMC

The Markov Chain Monte Carlo (MCMC) method we adopt samples from the complete sequence of conditional distributions to estimate the model parameters ρ, β, θ, and σ. Specifically, we sample from the conditional distribution $F(\beta^1|\rho^0, \sigma^0)$, where ρ^0, σ^0 represent arbitrary starting values and β^1 is a draw. The initial draw β^1 is then used when sampling from the conditional distribution $F(\sigma^1|\beta^1, \rho^0)$, and then from $F(\rho^1|\beta^1, \sigma^1)$. We then return to sample a new value for β^2 from $F(\beta^2|\rho^1, \sigma^1)$, followed by sampling from $F(\sigma^2|\beta^2, \rho^1)$ and $F(\rho^2|\beta^2, \sigma^2)$. This process continues iteratively, and a large sample of parameters drawn in this sequence are used to calculate posterior means and variances. These moments are equivalent to maximum likelihood estimates when no prior distributions are assigned to the parameters.

Given the large sample of firms in our sample, prior information would have minimal impact on the resulting parameter estimates and thus our estimates are not sensitive to the use of diffuse priors. The conditional distributions can be derived from the model likelihood function by considering the distribution for each parameter, assuming that all other parameters are known. For ease of notation, we write the explanatory variables matrix as: $Z = \begin{pmatrix} X & WX \end{pmatrix}$, and the associated parameters as: $\delta = \begin{pmatrix} \beta' & \theta' \end{pmatrix}'$. The likelihood is shown in Eq. (12) below, where $|A|$ denotes the determinant of the $N \times N$ matrix A.

$$\begin{aligned} f(Y; \delta, \rho, \sigma^2) &= |I_N - \rho W|(2\pi\sigma^2)^{-N/2}\exp(-\frac{e'e}{2\sigma^2}) \\ e &= Y\omega - Z\delta_d \\ Y &= \begin{pmatrix} Y & MY \end{pmatrix},\ \omega = \begin{pmatrix} 1 \\ -\rho \end{pmatrix} \\ \delta_d &= (Z'Z)^{-1}Z'Y\omega \end{aligned} \tag{12}$$

Given the flat prior, the conditional distribution for δ is multivariate normal with mean and variance-covariance shown in Eq. (13).

$$\begin{aligned} F(\delta|\sigma^2, \rho) &= Normal(\bar{\delta}, \bar{\Sigma}_\delta), \\ \bar{\delta} &= (Z'Z)^{-1}(Z'Y\omega), \\ \bar{\Sigma}_\delta &= \sigma^2(Z'Z)^{-1}. \end{aligned} \tag{13}$$

We note that $(Z'Z)^{-1}Z'Y$ consists of only sample data information, so this expression (and also $(Z'Z)^{-1}$) can be calculated once prior to MCMC sampling. Consequently, the sampling of new values of δ (given values for the parameters σ, ρ) can take place in a rapid and computationally efficient way.

The conditional posterior for σ^2 (given δ, ρ) given a flat prior takes the form in Eq. (14) below.

$$\begin{aligned} F(\sigma^2|\delta, \rho) &\propto (\sigma^2)^{-(\frac{N}{2})}\exp\left(-\frac{1}{2\sigma^2}(\omega'e'e\omega)\right) \\ e &= (Y - Z\delta_d) \\ \delta_d &= (Z'Z)^{-1}(Z'Y) \\ &\sim Inverse\ Gamma(a, b) \\ a &= N/2 \\ b &= (\omega'e'e\omega)/2 \end{aligned} \tag{14}$$

As before, the expression δ_d consists only of sample data, allowing this to be computed once prior to beginning the MCMC sampling loop.

The conditional distribution for the dependence parameter ρ does not take the form of a conventional distribution, so we rely on a method that is often labeled *draw-by-inversion* in the literature. We begin with the full joint posterior $F(\rho, \delta, \sigma^2)$ and analytically integrate out δ, σ^2. This relies on standard techniques from the Bayesian regression literature (e.g., Zellner, 1971). Combining the likelihood function in Eq. (12) with flat priors (and ignoring the constant $2\pi^{N/2}$) leads, after integrating out δ, to the joint posterior given in Eq. (16) below. Using $\Gamma()$ to denote the Gamma function and integrating out σ^2 then yields Eq. (17) below.

$$f(Y; \rho, \sigma^2, \delta) \propto \int |I_N - \rho W|(\sigma^2)^{-(N+2)/2}\exp(-\frac{1}{2\sigma^2}\omega'e'e\omega)d\delta \tag{15}$$

$$F(\rho, \sigma^2|Y) \propto |I_N - \rho W| \int_0^\infty \sigma^{-(N+2)}\exp\left(-\frac{1}{2\sigma^2}\omega'e'e\omega\right)d\sigma \tag{16}$$

$$F(\rho|Y) = 2^{(N-2)/2}\Gamma(N/2)|I_N - \rho W||Z'Z|^{-1/2}(\omega'e'e\omega)^{-(N-k)/2} \tag{17}$$

On each trip through the MCMC sampler, we use univariate numerical integration of Eq. (17) over the parameter ρ, which is constrained to lie in the interval $(-1 < \rho < 1)$.[27] A draw for ρ is produced by inversion of the cumulative probability density function arising from this integration in conjunction with a random deviate (see LeSage and Pace, 2009, Chapter 5 for additional details).

A computation challenge in the MCMC estimation involves calculating the determinant of the large $N \times N$ matrix $I_N - \rho W$. We can work with the log-joint posterior

[27] The open interval $(-1, 1)$ for ρ ensures that $(I_N - \rho W)^{-1}$ exists, so the data generating process and reduced form are well-defined.

Table A.1
Variable definitions. This table provide the definitions of all of our variables using Compustat/CRSP variable names. The definitions and timing conventions follow those of Leary and Roberts (2014) where applicable. Additional variables and timing conventions follow Grieser and Hadlock (2019). The prefix "l." indicates a one-year lag. All variables in the paper and before entry into these formulas are inflation adjusted to 1971 dollars.

Variable name	Definition/construction
Book leverage	(dltt+dlc)/at
Market leverage	(dltt+dlc)/(at + prccf × csho - ceq)
Debt issuance	(dltt + dlc - (l.dltt+l.dlc))/l.at
Equity issuance	(ceq - re - (l.ceq - l.re))/l.at
Capex	capx/l.at
R&D	xrd/l.at where xrd is set equal to zero if missing
Market-to-book	(at+(prccf × csho)-ceq)/at
ROA	ni/l.at
Log(sales)	log(sale)
Q	(at +(prccd × csho) - ceq)/at
Zscore	6.56 × wcap/at + 3.26 × re/at + 6.72 × ebit/at + 1.05 × ceq/lt
Log(mktcap)	log(prccf × csho)
Tangibility	ppent/at
Cash flow	(opiti+dp)/l.at
Cash	(che)/at
Age	current year - IPO year
Debtdue	dd1/(dltt + dlc)
Asset growth	at/l.at - 1
Dividend yield	dvc/prcc_f
Stock return	exp[sum over days in year[log(1+ mkt.-adj. daily ret)]] - 1

for $F(\rho|Y)$ which results in $log|(I_N - \rho W)| = \sum_{t=1}^{T} \log|I_{n_t} - \rho W_t|$. Since the matrices W_t are sparse (containing a great number of zero values), there are sparse Cholesky decomposition routines available in the MATLAB software that we use to rapidly calculate the T different (logged) determinants (see LeSage and Pace, 2009). We calculate these over a fine grid of values for the parameter ρ over the range $(-1, 1)$ and store the results prior to beginning MCMC sampling. Of note, $e'e$ can also be precalculated using the definition $e = (Y - Z\delta_d)$ which involves only sample data. Thus, we can evaluate $F(\rho|Y)$ over the grid of values for ρ, facilitating rapid univariate numerical integration using a trapezoid rule.

A key advantage of the MCMC procedure is that it produces an entire chain of parameter estimates, akin to a bootstrapping procedure. These estimates can be treated as samples from the true probability distribution of the parameters. Consequently, calculating confidence intervals around the mean (median) estimates from these samples is straightforward. For a large number of drawings, the standard error is simply the standard deviation of the sampled estimates. Confidence intervals and t-statistics can be derived from these standard errors with no further adjustment.

A1.5. Interpreting coefficients and direct/indirect effects

In models that allow for peer network effects, the cross-partial derivative of y_i with respect to x_j is potentially nonzero. If we define $S = (I_N - \rho W)^{-1}$, then in our framework the *indirect effect* (network effect) is defined as:

$$\frac{\partial y_i}{\partial x_j} = S_{ij}(\beta + w_{ij}\theta) \tag{18}$$

where S_{ij} is the row i, column j element of the matrix S. Kelejian and Piras (2017) and other authors characterize this effect as the emanating effect of unit j onto unit i. To the best of our knowledge, Leary and Roberts (2014) is the only peer-effect study in finance that grapples with this nuance of parameter interpretation.

A change in an explanatory variable at unit i can affect the outcome variable of peers, which can "feed back" to affect the outcome of unit i itself. This feedback loop implies that the own derivative $\partial y_i/\partial x_i$ will not generally equal β if ρ is nonzero. The *direct effect* is defined as:

$$\frac{\partial y_i}{\partial x_i} = S_{ii}\beta, \tag{19}$$

and the own spillover (feedback loop) effect of x_i on y_i can be defined as:

$$(S_{ii} - 1)\beta. \tag{20}$$

Eq. (19) and Eq. (20) once again illustrate the challenges of ignoring peer effects when they are present, but not of primary interest, as even own-firm effects will no longer have a simple linear partial derivative interpretation.

Eq. (18) and Eq. (19) are specific to each observation or observation pair. Thus, there are N^2 total partial derivatives. To calculate scalar summaries of the *direct* and *indirect* effects, we simply take the average for the full panel of firm-years. Specifically, we calculate the average direct effect as $\frac{1}{N}\sum_i (I - \rho W)^{-1}_{ii}\beta$ and the average indirect effect as $\frac{1}{N}\sum_i \sum_{j\neq i} (I - \rho W)^{-1}_{ij}(\beta + w_{ij}\theta)$.

A2. Simulation/bootstrap procedure

In Table 4 we report alternative standard errors derived by considering the distribution of the estimated ρ peer-effect coefficients under the null derived by using a set of uninformative peer-weighting matrices. To construct these matrices, in the first sample year, we identify a firm's weights on its (real) peers and instead assign these weights to a randomly selected set of other

Table A.2
Extended SDM models of peer effects in corporate leverage decisions. The table presents estimates of coefficients/effects for the extended Spatial Durbin (SDM) model outlined in the text and Appendix in which corporate leverage decisions are assumed to be a function of a weighted average of peer leverage, a selected exogenous explanatory variable, and a set of four additional control variables. The SDM model extends the SAR model to allow a firm's capital structure to directly depend on the peer explanatory variables. All other model, sample, and estimation details are the same as in the corresponding SAR model tabulated in Table 3.

	Market leverage	Book leverage	Equity issuance	Debt issuance	Δ Market leverage	Δ Book leverage
Panel A. Equity shock exogenous variable						
Peer-effect coefficient ρ	0.220***	0.186***	0.070***	0.060***	0.094***	0.030***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.004]
Direct effects						
Equity shock	−0.047***	−0.025***	0.104***	0.008*	−0.063***	−0.053***
	[0.004]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Log(sales)	0.159***	0.222***	−0.106***	0.145***	0.091***	0.080***
	[0.005]	[0.006]	[0.006]	[0.006]	[0.005]	[0.005]
Book-to-market	−0.281***	−0.028***	0.213***	−0.027***	−0.219***	−0.006
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]
Profit	−0.190***	−0.213***	−0.133***	−0.132***	−0.195***	−0.205***
	[0.005]	[0.006]	[0.006]	[0.006]	[0.005]	[0.005]
Tangibility	0.163***	0.185***	−0.007	0.094***	0.048***	0.071***
	[0.005]	[0.006]	[0.006]	[0.006]	[0.005]	[0.005]
Indirect effects						
Equity shock	0.005	0.006	0.007	−0.001	0.001	0.006
	[0.006]	[0.006]	[0.005]	[0.005]	[0.005]	[0.005]
Log(sales)	−0.064***	−0.063***	−0.045***	−0.023***	−0.010*	−0.012**
	[0.007]	[0.007]	[0.006]	[0.006]	[0.006]	[0.005]
Book-to-market	−0.093***	−0.053***	0.056***	−0.002	−0.021***	−0.008*
	[0.006]	[0.006]	[0.006]	[0.005]	[0.005]	[0.005]
Profit	−0.007	0.008	−0.016***	0.039***	−0.012**	−0.004
	[0.007]	[0.007]	[0.006]	[0.007]	[0.006]	[0.005]
Tangibility	0.016**	0.010	0.013**	−0.015**	0.030***	0.018***
	[0.007]	[0.007]	[0.006]	[0.007]	[0.005]	[0.005]
Panel B. Debt due exogenous variable						
Peer-effect coefficient ρ	0.265***	0.175***	0.141***	0.084***	0.159***	0.038***
	[0.006]	[0.006]	[0.006]	[0.006]	[0.006]	[0.007]
Direct effects						
Debt due	−0.169***	−0.202***	0.020***	−0.065***	−0.021***	−0.016***
	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]	[0.004]
Log(sales)	0.044***	0.082***	−0.164***	0.055***	0.059***	0.052***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]	[0.004]
Book-to-market	-0.288***	0.076***	0.162***	−0.013***	−0.196***	0.098***
	[0.004]	[0.004]	[0.005]	[0.005]	[0.004]	[0.004]
Profit	−0.188***	−0.278***	−0.033***	−0.058***	−0.168***	−0.326***
	[0.005]	[0.005]	[0.005]	[0.005]	[0.005]	[0.004]
Tangibility	0.121***	0.108***	−0.033***	0.066***	0.069***	0.061***
	[0.004]	[0.005]	[0.005]	[0.005]	[0.004]	[0.004]
Indirect effects						
Debt due	−0.061***	−0.065***	0.051***	−0.030***	−0.004	0.008
	[0.009]	[0.009]	[0.008]	[0.008]	[0.008]	[0.007]
Log(sales)	−0.010	−0.030***	−0.023**	−0.017**	0.002	−0.008
	[0.009]	[0.009]	[0.009]	[0.009]	[0.008]	[0.006]
Book-to-market	−0.116***	−0.047***	0.099***	0.008	−0.040***	0.009
	[0.010]	[0.008]	[0.009]	[0.008]	[0.009]	[0.008]
Profit	−0.038***	0.001	−0.049***	0.024**	-0.060***	-0.024**
	[0.011]	[0.010]	[0.010]	[0.010]	[0.010]	[0.010]
Tangibility	0.040***	0.022***	0.003	−0.013*	0.045***	−0.006
	[0.008]	[0.008]	[0.007]	[0.008]	[0.009]	[0.008]

Significance at * 10% level, ** 5% level, *** 1% level

firms (fake peers). In the subsequent year, if a selected peer is still in existence, we leave the weight and identity of the peer unchanged with a probability of 0.75 and replace the peer with an alternative randomly selected firm with a probability of 0.25 (leaving the weight unchanged). The 0.75/0.25 choice is selected to match the actual sample peer-relationship survival rate rounded to the nearest 0.05. If a selected peer is no longer in existence in a given year, we replace it with a randomly selected firm within the set of new entrant firms in that year and assign a random weight to that peer drawn from the empirical distribution of all weights. This process is carried forward through all years to create a set of weights which are then normalized so that all row sums are equal to 1. We repeat this

Table A.3

Specification choices for models of other corporate finance settings. This table outlines the independent variables selected for each of the noncapital structure dependent variable regressions reported in Table 9. Timing conventions follow Grieser and Hadlock (2019). Models of capital spending and R&D are taken directly from Grieser and Hadlock (2019), and the model of asset growth utilizes the same explanatory variables. The models of cash holdings and dividend policy use the same explanatory variables and timing conventions as in the Grieser and Hadlock (2019) models predicting financial choices.

Dependent variable	Independent variables						
Capex	*log(at)*	*Q*	*Cash flow*	*Cash*	*Zscore*	*Tangibility*	
R&D	*log(at)*	*Q*	*Cash flow*	*Cash*	*Zscore*	*Tangibility*	
Asset growth	*log(at)*	*Q*	*Cash flow*	*Cash*	*Zscore*	*Tangibility*	
Cash	*log(Sale)*	*Q*	*Age*	*ROA*	*Zscore*	*Tangibility*	*R&D*
Dividend yield	*log(Sale)*	*Q*	*Age*	*ROA*	*Zscore*	*Tangibility*	*Cash*

algorithmic procedure 100 times to create a set of uninformative peer matrices and estimate the model on each of these matrices. Reported standard errors and p-values under the null are then based on the empirical distribution of resulting model estimates.

A3. Variable selection and construction

All variables employed in the empirical analysis are selected to match as closely as possible the treatment in Leary and Roberts (2014) or Grieser and Hadlock (2019) and are inflation-adjusted and winsorized at the 1% and 99% tails. Table A.1 outlines the construction of each variable from underlying CRSP/Compustat data. In all models of capital structure, control variables that are stock values are measured at the same point in time as capital structure, while flow variables are measured for the fiscal year ending at the point in time that capital structure is measured. In the models of other dependent variables in Table 9, timing conventions are selected to match the treatment in Grieser and Hadlock (2019). The reported models of capital spending and R&D are taken directly from Grieser and Hadlock (2019), and the model of asset growth exactly parallels these models but with the annual growth in book assets as the dependent variable. The models of cash holdings and dividend policy use the same explanatory variables as in the Grieser and Hadlock (2019) models predicting financial choices. Table A.3 details the exact variables used in each Table 9 model.

The equity shock variable is constructed using the procedure outlined by LR and is a measure of a firm's annual industry/market-adjusted abnormal stock returns. We first estimate the regression:

$$R_{ijt} = \alpha_{ijt} + \beta^{M}_{ijt}(R^{M}_{t} - RF_t) + \beta^{IND}_{ijt}(\bar{R}_{-ijt} - RF_t) + \eta_{ijt},$$

where R_{ijt} is the return to firm i in industry j during month t, $R^M_t - RF_t$ is the excess return on the market, and $\bar{R}_{-ijt} - RF_t$ is the excess return on an equal-weighted industry portfolio excluding firm i's (industries defined by three-digit SIC). We require at least 24 monthly observations over the lagged 60-month period. The average coefficient estimates from these rolling stock-level regressions are $\alpha = 0.008$, $\beta^M = 0.419$, and $\beta^{IND} = 0.626$. Parameter estimates from this equation are used to predict monthly returns, and the equity shock is the resulting residual, with monthly shocks compounded to arrive at an annual figure. This annualized variable, lagged one year as in LR (i.e., for the period ending one year prior to the capital structure prediction date), is the explanatory variable used in the tabulated regressions.

References

Adhikari, B.K., Agrawal, A., 2018. Peer influence on payout policies. J. Corp. Financ. 48, 615–637.
Almazan, A., Molina, C., 2005. Intra-industry capital structure dispersion. J. Econ. Manag. Strategy 14, 263–297.
Almeida, H., Campello, M., Laranjeira, B., Weisbenner, S., 2012. Corporate debt maturity and the real effects of the 2007 credit crisis. Crit. Financ. Rev. 1 (1), 3–58.
Angrist, J.D., 2014. The perils of peer effects. Labour Econ. 30, 98–108.
Bailey, M., Cao, R., Kuchler, T., Stroebel, J., 2018. The economic effects of social networks: evidence from the housing market. J. Polit. Econ. 126 (6), 2224–2276.
Bailey, M., Johnston, D.M., Kuchler, T., Stroebel, J., Wong, A., 2019. Peer Effects in Product Adoption. Technical Report. National Bureau of Economic Research.
Bolton, P., Scharfstein, D.S., 1990. A theory of predation based on agency problems in financial contracting. Am. Econ. Rev. 80, 93–106.
Bradley, M., Jarrell, G.A., Kim, E.H., 1984. On the existence of an optimal capital structure: theory and evidence. J. Financ. 39 (3), 857–878.
Bramoullé, Y., Djebbari, H., Fortin, B., 2009. Identification of peer effects through social networks. J. Econ. 150 (1), 41–55.
Brander, J.A., Lewis, T.R., 1986. Oligopoly and financial structure: the limited liability effect. Am. Econ. Rev. 76, 956–970.
Bustamante, M.C., Frésard, L., 2020. Does firm investment respond to peers' investment? Robert H. Smith School Research Paper No. RHS 2827803.
Campello, M., 2003. Capital structure and product markets interactions: evidence from business cycles. J. Financ. Econ. 68 (3), 353–378.
Campello, M., 2006. Debt financing: does it boost or hurt firm performance in product markets? J. Financ. Econ. 82 (1), 135–172.
Cao, J., Liang, H., Zhan, X., 2019. Peer effects of corporate social responsibility. Manag. Sci. 65 (12), 5487–5503.
Case, A.C., Katz, L.F., 1991. The Company you Keep: the Effects of Family and Neighborhood on Disadvantaged Youths. Technical Report. National Bureau of Economic Research.
Chevalier, J.A., 1995. Capital structure and product-market competition: empirical evidence from the supermarket industry. Am. Econ. Rev. 85, 415–435.
Chevalier, J.A., Sharfstein, D.S., 1996. Capital market imperfections and countercyclical markups: theory and evidence. Am. Econ. Rev. 86 (4), 703–725.
Cookson, J.A., 2018. Anticipated entry and entry deterrence: evidence from the american casino industry. Manag. Sci. 64 (5), 2325–2344.
Cookson, J. A., Engelberg, J., Mullins, W., 2021. Echo chambers. Unpublished working paper. SSRN.
Dasgupta, S., Titman, S., 1998. Pricing strategy and financial policy. Rev. Financ. Stud. 11 (4), 705–737.
Dougal, C., Parsons, C.A., Titman, S., 2015. Urban vibrancy and corporate growth. J. Financ. 70 (1), 163–210.
Duflo, E., Saez, E., 2003. The role of information and social interactions in retirement plan decisions: evidence from a randomized experiment. Q. J. Econ. 118 (3), 815–842.

Fairhurst, D., Nam, Y., 2020. Corporate governance and financial peer effects. Financ. Manag. 49 (1), 235–263.
Fracassi, C., 2016. Corporate finance policies and social networks. Manag. Sci. 63 (8), 2420–2438.
Frank, M.Z., Goyal, V.K., 2009. Capital structure decisions: which factors are reliably important? Financ. Manag. 38 (1), 1–37.
Grennan, J., 2019. Dividend payments as a response to peer influence. J. Financ. Econ. 131 (3), 549–570.
Grieser, W.D., Hadlock, C.J., 2019. Panel-data estimation in finance: testable assumptions and parameter (in) consistency. J. Financ. Quant. Anal. 54 (1), 1–29.
Heimer, R.Z., 2016. Peer pressure: social interaction and the disposition effect. Rev. Financ. Stud. 29 (11), 3177–3209.
Hoberg, G., Phillips, G., 2010. Product market synergies and competition in mergers and acquisitions: a text-based analysis. Rev. Financ. Stud. 23 (10), 3773–3811.
Hoberg, G., Phillips, G., 2016. Text-based network industries and endogenous product differentiation. J. Polit. Econ. 124 (5), 1423–1465.
Kaustia, M., Rantala, V., 2015. Social learning and corporate peer effects. J. Financ. Econ. 117 (3), 653–669.
Kelejian, H., Piras, G., 2017. Spatial Econometrics. Academic Press, London.
Khanna, N., Tice, S., 2000. Strategic responses of incumbents to new entry: the effect of ownership structure, capital structure, and focus. Rev. Financ. Stud. 13 (3), 749–779.
Khanna, N., Tice, S., 2005. Pricing, exit, and location decisions of firms: evidence on the role of debt and operating efficiency. J. Financ. Econ. 75 (2), 397–427.
Kovenock, D., Phillips, G.M., 1997. Capital structure and product market behavior: an examination of plant exit and investment decisions. Rev. Financ. Stud. 10 (3), 767–803.
Kuchler, T., Russel, D., Stroebel, J., 2020. The Geographic Spread of COVID-19 Correlates with the Structure of Social Networks as Measured by Facebook. Technical Report. National Bureau of Economic Research.
Leary, M.T., Roberts, M.R., 2014. Do peer firms affect corporate financial policy? J. Financ. 69 (1), 139–178 .
Lee, L.-f., 2007. Identification and estimation of econometric models with group interactions, contextual factors and fixed effects. J. Econ. 140 (2), 333–374.
LeSage, J.P., Pace, R.K., 2009. Introduction to Spatial Econometrics. Chapman and Hall/CRC, New York.
MacKay, P., Phillips, G.M., 2005. How does industry affect firm financial structure? Rev. Financ. Stud. 18 (4), 1433–1466.
Maksimovic, V., 1988. Capital structure in repeated oligopolies. Rand J. Econ. 19, 389–407.
Maksimovic, V., 1995. Financial structure and product market competition. In: Handbooks in Operations Research and Management Science, 9, pp. 887–920.
Manski, C.F., 1993. Identification of endogenous social effects: the reflection problem. Rev. Econ. Stud. 60 (3), 531–542.
Parsons, C., Titman, S., 2008. Capital Structure and Corporate Strategy. In: *Handbook of Empirical Corporate Finance*. Elsevier, San Diego, pp. 203–234.
Phillips, G.M., 1995. Increased debt and industry product markets an empirical analysis. J. Financ. Econ. 37 (2), 189–238.
Scharfstein, D.S., Stein, 1990. Herd behavior and investment. Am. Econ. Rev. 80 (3), 465–479.
Showalter, D.M., 1995. Oligopoly and financial structure: comment. Am. Econ. Rev. 85 (3), 647–653.
Shue, K., 2013. Executive networks and firm policies: evidence from the random assignment of MBA peers. Rev. Financ. Stud. 26 (6), 1401–1442.
Silva, A.F., 2019. Strategic liquidity mismatch and financial sector stability. Rev. Financ. Stud. 32 (12), 4696–4733.
Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. Nat. Hum. Behav. 4 (11), 1208–1214.
Titman, S., Wessels, R., 1988. The determinants of capital structure choice. J. Financ. 43 (1), 1–19.
Zellner, A., 1971. Bayesian and non-bayesian analysis of the log-normal distribution and log-normal regression. J. Am. Stat. Assoc. 66 (334), 327–330.
Zingales, L., 1998. Survival of the fittest or the fattest? exit and financing in the trucking industry. J. Financ. 53 (3), 905–938.